

Supporting Information

Ketazine-Linked Crystalline Porous Covalent Organic Frameworks

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Contents

Section 1. Methods	2
Section 2. Materials and Synthetic Procedures	5
Section 3. Supporting Tables	7
Section 4. Supporting Figures	28
Section 5. Supporting References	56

Section 1. Methods

Characterization

Powder X-Ray diffraction (PXRD) patterns were recorded on Bruker D8 ADVANCE diffractometer with a Cu/K α anode ($\lambda = 1.5406 \text{ \AA}$) by depositing powder on poly(methyl methacrylate) (PMMA) substrate, from 2° to 30° at a 0.01° interval.

Nitrogen sorption experiments were performed at 77 K on Micromeritics Instrument Corporation model 3Flex surface characterization. The specific surface areas were calculated using Brunauer–Emmett–Teller (BET) method. In order to estimate pore size distributions, non-local density functional theory (NLDFT) model was adopted to analyze the nitrogen isotherms on the basis of the model of on carbon with cylinder pores, while the pore volume was derived from the sorption curve. Before the adsorption test, powder samples were heated at 120°C for 12 h.

Fourier-transform infrared (FT IR) spectroscopy was performed on a Bruker ALPHA FT IR model OPUS infrared spectrophotometer with KBr disks in a frequency range of $4000 - 500 \text{ cm}^{-1}$.

^1H NMR spectra were recorded on a Bruker 500 MHz NMR spectrometer. Solid state ^{13}C cross polarization magic angle spinning nuclear magnetic resonance spectroscopy (^{13}C CP/MAS NMR) spectroscopy were carried out on a 14.1 T Bruker ACNEO400 MHz spectrometer, using a 1.9 mm Bruker HXY probe at an MAS frequency of 24 kHz.

Thermogravimetric analysis (TGA) was conducted on a TA Instruments model Q500 by heating the samples from 40 to 800°C at a rate of $10^\circ\text{C min}^{-1}$ under nitrogen flow (10 mL min^{-1}).

CHN Elemental analysis was carried out on a ThermoFisher Scientific FlashSmart CHNS Elemental Analyzer by weighing the samples in tin containers and then combusted at high temperature with oxygen.

Field emission scanning electron microscopy (FE SEM) was performed on a JEOL model JSM6701F field emission scanning electron microscope at an accelerating voltage of 5 kV at a working distance of 8.0 mm. The COF samples were dispersed in ethanol, sonicated for 60 sec, drop-casted onto a silicon wafer, and coated with Pt with a sputter coater at ambient temperature (298 K) under vacuum.

High resolution transmission electron microscopy (HR TEM) images of COF were acquired from a FEI-Titan (Themis ETEM, operated at 300 kV) under a low dose mode. The sample was prepared by drop-casting a supersonicated ethanol suspension of COFs onto a copper grid.

The electronic absorption spectroscopy was performed on a JASCO model V-770 spectrophotometer. Solid-state electronic absorption spectra were carried out on a JASCO model V-770 spectrophotometer by measuring the reflectance of powders in the solid state, and further automatically converted from absorbance into Kubelka-Munk coefficient.

For solution measurements, the COF sample (1 mg) was dispersed in 5 mL of DCM, THF, DMF or NMP, respectively. After sonicating with a probe sonicator for 10 min, the homogeneous suspensions of COFs were obtained. The dispersion was kept undisturbed overnight to yield a transparent dispersion. The supernatant was collected with a pipette and diluted with corresponding solvent to a maximum absorbance (*Abs*) at the range of 0.1 – 0.3. For luminescence measurements, the supernatant was collected with a pipette and diluted with corresponding solvent to an absorbance (*Abs*) at the excitation wavelength less than 0.15, to eliminate the concentration effect on light emission.

Photoluminescence spectroscopy was carried out on a JASCO FP-8600 spectrophotometer with excitation at their absorption bands. The JASCO FP-8600 was equipped with a 60-mm integrating sphere (ILF-135, JASCO) and used for absolute fluorescence quantum yield measurements.

The fluorescence decay curves were conducted on a Hamamatsu model Quantaurus-Tau C16361-02 that contains a light-emitting diode (LED) pulsed light source, monochromator, PMT (photomultiplier tube), iris (aperture) for adjusting light level, and filter for cutting excitation light. The COF samples were irradiated with internal LED at 405 nm. The powder sample (1 mg) was dispersed in solvents and the resultant mixture was ultrasonicated for 60 min. The dispersion was kept undisturbed overnight to yield a transparent dispersion. The supernatant was collected with a pipette and diluted with corresponding solvent to an absorbance (*Abs*) at the excitation wavelength less than 0.1, to eliminate the concentration effect on light re-absorption.

Chemical Stability Test

The as-synthesized ketazine-COFs (20 mg) were kept in different solvents (4 mL), including THF, DMF, MeOH, boiling water, aqueous HCl (6 M) solution, and aqueous NaOH (6 M) solution for one week. The samples were washed with THF (for samples treated in organic solvents) or water (for samples treated in aqueous solutions), dried under vacuum at 85 °C for 12 h, and subjected to PXRD, FT IR, and weight analyses.

Computational methods

Molecular modelling and Pawley refinement were carried out using Reflex, a software package for crystal determination from PXRD pattern, implemented in Materials Studio 2020 (Accelrys Inc.). We performed Pawley refinement to optimize the lattice parameters iteratively until the R_{wp} and R_p values converge. The Thompson-Cox-Hastings profile function for whole profile fitting and asymmetry correction function as Berar-Baldinozzi were used during the refinement processes. The crystalline structures of COFs were determined using the density-functional tight-binding (DFTB+) method including Lennard-Jones (LJ) dispersion. DFTB+ is an approximate density functional theory method based on the empirical tight-binding approach, utilizing an optimized minimal LCAO Slater-type all-valence basis set in combination with a two-center approximation for Hamiltonian matrix elements. The lattice dimensions were optimized simultaneously with the geometry. Standard DFTB parameters for X-Y element pair (X, Y = C, H, and N) interactions were employed from the mio-0-1 set.

Section 2. Materials and Synthetic Procedures

Materials

Dichloromethane (DCM) and hexane were purchased from Avantor Performance Materials. Tetrahydrofuran (THF, stabilized with BHT), methanol (MeOH), and dimethylformamide (DMF) were purchased from VWR Chemicals. *o*-Dichlorobenzene (*o*-DCB), *n*-butanol (*n*-BuOH), and 1,4-dioxane were purchased from Sigma Aldrich. Mesitylene, anisole, benzonitrile, *m*-xylene, trifluoroacetic acid (TFA), and acetic acid (HOAc) were purchased from TCI Chemicals. 1,1',1''-(Benzene-1,3,5-triyl)triethanone (TKB), benzene-1,3,5-triyltrimethanol, 4-acetylphenylboronic acid, 4-formylphenylboronic acid, potassium bromide crystal block (KBr), 4,4''-dibromo-5'-(4-bromophenyl)-1,1':3',1''-terphenyl, and tetrakis(triphenylphosphine)palladium(0) were purchased from BLD Pharm. Potassium carbonate (K₂CO₃), sodium sulfate (Na₂SO₄), sodium hydroxide (NaOH), and hydrochloric acid (HCl) were purchased from Goodrich Chemical Enterprise. Chloroform-d³ (CDCl₃) was purchased from CIL incorporated.

Synthesis of TKBPB

TKBPB monomer was synthesized according to modified literature method^{S1}. 4,4''-Dibromo-5'-(4-bromophenyl)-1,1':3',1''-terphenyl (0.99 g, 1.83 mmol), 4-acetylphenylboronic acid (1.35 g, 8.24 mmol), potassium carbonate (1.90 g, 8.24 mmol), and tetrakis(triphenylphosphine) palladium (0) (127 mg, 0.11 mmol) were added to a two-necked round bottle flask (250 mL) equipped with a stir bar and condenser. The flask was evacuated and backfilled with argon three times. Dioxane (70 mL) and H₂O (7 mL) were mixed and added to the flask. The reaction mixture was heated to 110 °C for 72 h. After cooled at room temperature, the resulting mixture was evaporated, and the solid part was extracted with CHCl₃ (150 mL × 3). The combined CHCl₃ layer was washed with water (200 mL × 3) and brine (200 mL × 3) and dried over anhydrous MgSO₄. The CHCl₃ solution was filtered and CHCl₃ was evaporated to give a solid, which was subjected to column chromatography (silica gel, DCM/Hexane = 5/1 v/v) to yield TKBPB as a light grey solid in 72% yield. ¹H NMR (500 MHz, CDCl₃): δ (ppm) 8.08 (d, J = 8.5 Hz, 6H), 7.91 (s, 3H), 7.85 (d, J = 8.5 Hz, 6H), 7.81-7.74 (m, 12H), 2.67 (s, 9H).

Synthesis of TKBPB-Ketazine-COF

A Pyrex tube (10 mL) filled with TKBPB (19.8 mg, 30 μ mol), hydrazine (2.2 μ L, 45 μ mol) in a mixture of anisole/*n*-BuOH/pure AcOH (80/20/6 vol.; 2.12 mL) was degassed by three freeze–pump–thaw cycles. The tube was sealed and heated at 120 °C for 5 days. The precipitate was collected by suction filtration, washed five times with THF, and extracted via Soxhlet with THF overnight. The powder was dried at 120 °C under vacuum overnight to afford TKBPB-Ketazine-COF in an isolated yield of 78%.

Synthesis of TKB-Ketazine-COF

A Pyrex tube (10 mL) filled with TKB (20 mg, 98 μ mol), hydrazine (7.1 μ L, 45 μ mol), and acetophenone (34.3 μ L, 294 μ mol) in a mixture of anisole/*n*-BuOH/pure TFA (75/25/6 vol.; 2.12 mL) was degassed by three freeze–pump–thaw cycles. The tube was sealed and heated at 120 °C under vacuum for 5 days. The precipitate was collected by suction filtration, washed five times with THF, and extracted via Soxhlet with THF overnight. The powder was dried at 120 °C under vacuum overnight to afford TKB-Ketazine-COF in an isolated yield of 73%.

Section 4. Supporting Tables

Table S1. Effect of Solvent. Polymerization of TKBPB with hydrazine.

No	Solvent	Catalyst	Temperature (°C)	Time (days)	Yield (%)	PXRD
1	<i>o</i> -DCB (1mL)	AcOH (6 M, 100 µL)	120	5	No solid	-
2	dioxane (1mL)	AcOH (6 M, 100 µL)	120	5	No solid	-
3	<i>m</i> -xylene (1mL)	AcOH (6 M, 100 µL)	120	5	No solid	-
4	benzonitrile (1mL)	AcOH (6 M, 100 µL)	120	5	No solid	-
5	anisole (1mL)	AcOH (6 M, 100 µL)	120	5	No solid	-
6	<i>o</i> -DCB/ <i>n</i> -BuOH (1/1, 1mL)	AcOH (6 M, 100 µL)	120	5	No solid	-
7	<i>m</i> -xylene/ <i>n</i> -BuOH (1/1, 1mL)	AcOH (6 M, 100 µL)	120	5	No solid	-
8	benzonitrile/ <i>n</i> -BuOH (1/1, 1mL)	AcOH (6 M, 100 µL)	120	5	No solid	-
9	anisole/ <i>n</i> -BuOH (1/1, 1mL)	AcOH (6 M, 100 µL)	120	5	No solid	-
10	mesitylene/dioxane (1/1, 1 mL)	AcOH (6 M, 100 µL)	120	5	No solid	-

Table S2. Effect of Catalyst. Polymerization of TKBPB with hydrazine.

No	Solvent	Catalyst	Temperature (°C)	Time (days)	Yield (%)	PXRD
1	<i>o</i> -DCB/ <i>n</i> -BuOH (1/1, 1 mL)	AcOH (6 M, 100 µL)	120	5	No solid	-
2	mesitylene/dioxane (1/1, 1 mL)	AcOH (6 M, 100 µL)	120	5	No solid	-
3	<i>o</i> -DCB/ <i>n</i> -BuOH (1/1, 1 mL)	AcOH (9 M, 80 µL)	120	5	No solid	-
4	mesitylene/dioxane (1/1, 1 mL)	AcOH (9 M, 80 µL)	120	5	No solid	-
5	<i>o</i> -DCB/ <i>n</i> -BuOH (1/1, 1 mL)	AcOH (12 M, 60 µL)	120	5	No solid	-
6	mesitylene/dioxane (1/1, 1 mL)	AcOH (12 M, 60 µL)	120	5	No solid	-
7	<i>o</i> -DCB/ <i>n</i> -BuOH (1/1, 1 mL)	AcOH (pure, 50 µL)	120	5	68%	Medium
8	mesitylene/dioxane (1/1, 1 mL)	AcOH (pure, 50 µL)	120	5	72%	Medium
9	dioxane (1 mL)	AcOH (pure, 50 µL)	120	5	62%	Weak
10	<i>o</i> -DCB (1 mL)	AcOH (pure, 50 µL)	120	5	77%	Weak
11	anisole (1 mL)	AcOH (pure, 50 µL)	120	5	73%	Weak
12	anisole/ <i>n</i> -BuOH (1/1, 1 mL)	AcOH (pure, 50 µL)	120	5	80%	Strong

Table S3. Effect of Amount of Catalyst. Polymerization of TKBPB with hydrazine.

No	Solvent	Catalyst	Temperature (°C)	Time (days)	Yield (%)	PXRD
1	anisole/ <i>n</i> -BuOH (1/1, 1mL)	AcOH (pure, 30 µL)	120	5	75	Medium
2	anisole/ <i>n</i> -BuOH (1/1, 1mL)	AcOH (pure, 40 µL)	120	5	77	Medium
3	anisole/ <i>n</i> -BuOH (1/1, 1mL)	AcOH (pure, 50 µL)	120	5	84	Strong
4	anisole/ <i>n</i> -BuOH (1/1, 1mL)	AcOH (pure, 60 µL)	120	5	81	Strong
5	anisole/ <i>n</i> -BuOH (1/1, 1mL)	AcOH (pure, 70 µL)	120	5	72	Medium
6	anisole/ <i>n</i> -BuOH (1/1, 1mL)	AcOH (pure, 80 µL)	120	5	65	Medium

Table S4. Effect of Ratio of Solvents. Polymerization of TKBPB with hydrazine.

No	Solvent	Catalyst	Temperature (°C)	Time (days)	Yield (%)	PXRD
1	anisole/ <i>n</i> -BuOH (6/1, 1mL)	AcOH (pure, 60 μ L)	120	5d	74%	Strong
2	anisole/ <i>n</i> -BuOH (4/1, 1mL)	AcOH pure, 60 μ L)	120	5d	82%	Strong
3	anisole/ <i>n</i> -BuOH (3/1, 1mL)	AcOH (pure, 60 μ L)	120	5d	80%	Strong
4	anisole/ <i>n</i> -BuOH (2/1, 1mL)	AcOH (pure, 60 μ L)	120	5d	77%	Strong
5	anisole/ <i>n</i> -BuOH (1/1, 1mL)	AcOH (pure, 60 μ L)	120	5d	86%	Weak
6	anisole/ <i>n</i> -BuOH (1/2, 1mL)	AcOH (pure, 60 μ L)	120	5d	90%	No Peak
7	anisole/ <i>n</i> -BuOH (1/3, 1mL)	AcOH (pure, 60 μ L)	120	5d	92%	No Peak
8	anisole/ <i>n</i> -BuOH (1/4, 1mL)	AcOH (pure, 60 μ L)	120	5d	87%	No Peak
9	anisole/ <i>n</i> -BuOH (1/6, 1mL)	AcOH (pure, 60 μ L)	120	5d	89%	No Peak

Table S5. Effect of Temperature. Polymerization of TKBPB with hydrazine.

No	Solvent	Catalyst	Temperature (°C)	Time (days)	Yield (%)	PXRD
1	anisole/ <i>n</i> -BuOH (4/1, 1mL)	AcOH (pure, 60 µL)	90	5	68	Weak
2	anisole/ <i>n</i> -BuOH (4/1, 1mL)	AcOH (pure, 60 µL)	120	5	82	Strong
3	anisole/ <i>n</i> -BuOH (4/1, 1mL)	AcOH (pure, 60 µL)	150	5	57	Weak

Table S6. Effect of Ratio of Reaction Time. Polymerization of TKBPB with hydrazine.

No	Solvent	Catalyst	Temperature (°C)	Time (days)	Yield (%)	PXRD
1	anisole/ <i>n</i> -BuOH (4/1, 1mL)	AcOH (pure, 60 µL)	120	1	75%	Weak
2	anisole/ <i>n</i> -BuOH (4/1, 1mL)	AcOH (pure, 60 µL)	120	2	81%	Medium
3	anisole/ <i>n</i> -BuOH (4/1, 1mL)	AcOH (pure, 60 µL)	120	3	83%	Strong
4	anisole/ <i>n</i> -BuOH (4/1, 1mL)	AcOH (pure, 60 µL)	120	4	80%	Strong
5	anisole/ <i>n</i> -BuOH (4/1, 1mL)	AcOH (pure, 60 µL)	120	5	82%	Strong
6	anisole/ <i>n</i> -BuOH (4/1, 1mL)	AcOH (pure, 60 µL)	120	6	80%	Strong
7	anisole/ <i>n</i> -BuOH (4/1, 1mL)	AcOH (pure, 60 µL)	120	7	81%	Strong

Table S7. Effect of trifluoroacetic acid catalyst. Polymerization of TKBPB with hydrazine.

No	Solvent	Catalyst	Temperature (°C)	Time (days)	Yield (%)	PXRD
1 (Control)	anisole/ <i>n</i> -BuOH (4/1, 1mL)	AcOH (pure, 60 µL)	120	5	82	Strong
2	anisole/ <i>n</i> -BuOH (4/1, 1mL)	TFA (0.5 M, 70 µL)	120	5	71	Medium
3	anisole/ <i>n</i> -BuOH (4/1, 1mL)	TFA (0.5 M, 90 µL)	120	5	73	Medium
4	anisole/ <i>n</i> -BuOH (4/1, 1mL)	TFA (pure, 30 µL)	120	5	68	Weak
5	anisole/ <i>n</i> -BuOH (4/1, 1mL)	TFA (pure, 40 µL)	120	5	67	Weak
6	anisole/ <i>n</i> -BuOH (4/1, 1mL)	TFA (pure, 50 µL)	120	5	72	Weak
7	anisole/ <i>n</i> -BuOH (4/1, 1mL)	TFA (pure, 60 µL)	120	5	77	Weak
8	anisole/ <i>n</i> -BuOH (4/1, 1mL)	TFA (pure, 70 µL)	120	5	71	Weak

Table S8. Effect of Solvents. Polymerization of TKB with hydrazine.

No	Solvent	Catalyst	Temperature (°C)	Time (days)	Yield (%)	XRD results
1	mesitylene/dioxane (1/1, 1 mL)	AcOH (6 M, 100 µL)	120	5	No solid	-
2	dioxane (1mL)	AcOH (6 M, 100 µL)	120	5	No solid	-
3	<i>o</i> -DCB (1mL)	AcOH (6 M, 100 µL)	120	5	No solid	-
4	<i>m</i> -xylene (1mL)	AcOH (6 M, 100 µL)	120	5	No solid	-
5	benzonitrile (1mL)	AcOH (6 M, 100 µL)	120	5	No solid	-
6	anisole (1mL)	AcOH (6 M, 100 µL)	120	5	No solid	-
7	<i>o</i> -DCB/ <i>n</i> -BuOH (1/1, 1 mL)	AcOH (6 M, 100 µL)	120	5	No solid	-
8	<i>m</i> -xylene/ <i>n</i> -BuOH (3/1, 1 mL)	AcOH (6 M, 100 µL)	120	5	No solid	-
9	benzonitrile/ <i>n</i> -BuOH (1/1, 1 mL)	AcOH (6 M, 100 µL)	120	5	No solid	-
10	anisole/ <i>n</i> -BuOH (3/1, 1 mL)	AcOH (6 M, 100 µL)	120	5	No solid	-
11	mesitylene/dioxane (1/1, 1 mL)	AcOH (9 M, 80 µL)	120	5	No solid	-
12	dioxane (1mL)	AcOH (9 M, 80 µL)	120	5	No solid	-
13	<i>o</i> -DCB (1mL)	AcOH (9 M, 80 µL)	120	5	No solid	-
14	<i>m</i> -xylene (1mL)	AcOH (9 M, 80 µL)	120	5	No solid	-
15	benzonitrile (1mL)	AcOH (9 M, 80 µL)	120	5	No solid	-
16	anisole (1mL)	AcOH (9 M, 80 µL)	120	5	No solid	-
17	<i>o</i> -DCB/ <i>n</i> -BuOH (1/1, 1 mL)	AcOH (9 M, 80 µL)	120	5	No solid	-
18	<i>m</i> -xylene/ <i>n</i> -BuOH (3/1, 1 mL)	AcOH (9 M, 80 µL)	120	5	No solid	-
19	Benzo/ <i>n</i> -BuOH (1/1, 1 mL)	AcOH (9 M, 80 µL)	120	5	No solid	-
20	anisole/ <i>n</i> -BuOH (3/1, 1 mL)	AcOH (9 M, 80 µL)	120	5	No solid	-
21	mesitylene/dioxane (1/1, 1 mL)	AcOH (pure, 50 µL)	120	5	gel	-
22	dioxane (1mL)	AcOH (pure, 50 µL)	120	5	gel	-
23	<i>o</i> -DCB (1mL)	AcOH (pure, 50 µL)	120	5	gel	-
24	<i>m</i> -xylene	AcOH	120	5	gel	-

	(1mL)	(pure, 50 μ L)				
25	benzonitrile (1mL)	AcOH (pure, 50 μ L)	120	5	No solid	-
26	anisole (1mL)	AcOH (pure, 50 μ L)	120	5	gel	-
27	<i>o</i> -DCB/ <i>n</i> -BuOH (1/1, 1 mL)	AcOH (pure, 50 μ L)	120	5	gel	-
28	<i>m</i> -xylene/ <i>n</i> -BuOH (3/1, 1 mL)	AcOH (pure, 50 μ L)	120	5	gel	-
29	benzonitrile/ <i>n</i> -BuOH (1/1, 1 mL)	AcOH (pure, 50 μ L)	120	5	No solid	-
30	anisole/ <i>n</i> -BuOH (3/1, 1 mL)	AcOH (pure, 50 μ L)	120	5	gel	-
31	anisole/ <i>n</i> -BuOH (3/1, 1 mL)	AcOH (pure, 50 μ L)	90	5	No solid	-
32	anisole/ <i>n</i> -BuOH (3/1, 1 mL)	AcOH (pure, 50 μ L)	150	5	No solid	-

Table S9. Effect of Acid. Polymerization of TKB with hydrazine.

No	Solvent	Catalyst	Temperature (°C)	Time (days)	Yield (%)	XRD results
1	anisole/ <i>n</i> -BuOH (3/1, 1 mL)	PTSA (0.5 M 50 µL)	120	5	No solid	-
2	anisole/ <i>n</i> -BuOH (3/1, 1 mL)	PTSA (0.5 M 70 µL)	120	5	No solid	-
3	anisole/ <i>n</i> -BuOH (3/1, 1 mL)	TFA (0.5 M 50 µL)	120	5	No solid	-
4	anisole/ <i>n</i> -BuOH (3/1, 1 mL)	TFA (0.5 M 70 µL)	120	5	No solid	-
5	anisole/ <i>n</i> -BuOH (3/1, 1 mL)	TFA (pure, 30 µL)	120	5	77%	Medium
6	anisole/ <i>n</i> -BuOH (3/1, 1 mL)	TFA (pure, 40 µL)	120	5	72%	Medium
7	anisole/ <i>n</i> -BuOH (3/1, 1 mL)	TFA (pure, 50 µL)	120	5	76%	Strong
8	anisole/ <i>n</i> -BuOH (3/1, 1 mL)	TFA (pure, 60 µL)	120	5	82%	Strong
9	anisole/ <i>n</i> -BuOH (3/1, 1 mL)	TFA (pure, 70 µL)	120	5	75%	Medium
10	anisole/ <i>n</i> -BuOH (3/1, 1 mL)	TFA (pure, 80 µL)	120	5	65%	Medium

Table S10. Effect of Amount of Modulator. Polymerization of TKB with hydrazine.

No	Solvent	Acetophenone	Catalyst	Temperature (°C)	Time (days)	Yield (%)	XRD results
1	anisole/ <i>n</i> -BuOH (3/1, 1 mL)	1eq.	TFA (pure, 60 µL)	120	5	78%	Weak
2	anisole/ <i>n</i> -BuOH (3/1, 1 mL)	2eq.	TFA (pure, 60 µL)	120	5	72%	Medium
3	anisole/ <i>n</i> -BuOH (3/1, 1 mL)	3eq.	TFA (pure, 60 µL)	120	5	82%	Strong
4	anisole/ <i>n</i> -BuOH (3/1, 1 mL)	4eq.	TFA (pure, 60 µL)	120	5	67%	Weak
5	anisole/ <i>n</i> -BuOH (3/1, 1 mL)	5eq.	TFA (pure, 60 µL)	120	5	54%	Weak
6	anisole/ <i>n</i> -BuOH (3/1, 1 mL)	6eq.	TFA (pure, 60 µL)	120	5	No solid	-

Table S11. Effect of Solvent, Temperature, and Reaction Time. Polymerization of TKB with hydrazine.

No	Solvent	Catalyst	Temp.	Time	Yield/%	XRD results
1	<i>o</i> -DCB/ <i>n</i> -BuOH (1/1, 1 mL)	TFA (pure, 60 μ L)	120	5	78%	Medium
2	<i>m</i> -xylene/ <i>n</i> -BuOH (3/1, 1 mL)	TFA (pure, 60 μ L)	120	5	72%	Medium
3	anisole/ <i>n</i> -BuOH (3/1, 1 mL)	TFA (pure, 60 μ L)	120	5	86%	Strong
4	mesitylene/dioxane (1/1, 1 mL)	TFA (pure, 60 μ L)	120	5	45%	Weak
5	anisole (1mL)	TFA (pure, 60 μ L)	120	5	-	NP
6	<i>o</i> -DCB (1mL)	TFA (pure, 60 μ L)	120	5	-	NP
8	anisole/ <i>n</i> -BuOH (6/1, 1 mL)	TFA (pure, 60 μ L)	120	5	62%	Weak
9	anisole/ <i>n</i> -BuOH (4/1, 1 mL)	TFA (pure, 60 μ L)	120	5	71%	Strong
10	anisole/ <i>n</i> -BuOH (3/1, 1 mL)	TFA (pure, 60 μ L)	120	5	82%	Strong
11	anisole/ <i>n</i> -BuOH (2/1, 1 mL)	TFA (pure, 60 μ L)	120	5	76%	Medium
12	anisole/ <i>n</i> -BuOH (1/1, 1 mL)	TFA (pure, 60 μ L)	120	5	77%	NP
13	anisole/ <i>n</i> -BuOH (1/2, 1 mL)	TFA pure, 60 μ L)	120	5	82%	NP
14	anisole/ <i>n</i> -BuOH (1/3, 1 mL)	TFA (pure, 60 μ L)	120	5	84%	NP
15	anisole/ <i>n</i> -BuOH (1/4, 1 mL)	TFA (pure, 60 μ L)	120	5	78%	NP
16	anisole/ <i>n</i> -BuOH (1/6, 1 mL)	TFA (pure, 60 μ L)	120	5	85%	NP
18	anisole/ <i>n</i> -BuOH (3/1, 1 mL)	TFA (pure, 60 μ L)	90	5	64%	Weak
19	anisole/ <i>n</i> -BuOH (3/1, 1 mL)	TFA (pure, 60 μ L)	150	5	No solid	NP
20	anisole/ <i>n</i> -BuOH (3/1, 1 mL)	TFA (pure, 60 μ L)	120	1	38%	Weak
21	anisole/ <i>n</i> -BuOH (3/1, 1 mL)	TFA (pure, 60 μ L)	120	2	56%	Medium
22	anisole/ <i>n</i> -BuOH (3/1, 1 mL)	TFA (pure, 60 μ L)	120	3	76%	Strong
23	anisole/ <i>n</i> -BuOH (3/1, 1 mL)	TFA (pure, 60 μ L)	120	4	74%	Strong
24	anisole/ <i>n</i> -BuOH (3/1, 1 mL)	TFA (pure, 60 μ L)	120	5	82%	Strong
25	anisole/ <i>n</i> -BuOH	TFA	120	6	78%	Strong

	(3/1, 1 mL)	pure, 60 μ L)				
26	anisole/ <i>n</i> -BuOH (3/1, 1 mL)	TFA (pure, 60 μ L)	120	7	81%	Strong

Table S12. Elemental analysis. Elemental analysis of COFs.

COF		C%	H%	N%
TKB-Ketazine-COF	Calculated	72.73	6.06	21.21
	Observed	72.18	6.01	19.84
TKBPB-Ketazine-COF	Calculated	88.07	5.50	6.43
	Observed	86.02	5.44	6.14

Table S13. Atomic coordinates. P6 space group coordinates of TKB-Ketazine-COF. $a = b = 15.2353 \text{ \AA}$, $c = 4.0243 \text{ \AA}$, $\alpha = \beta = 90^\circ$, and $\gamma = 120^\circ$.

Atom Name	Atom	x	y	z
C1	C	0.23023	-0.41085	-0.48157
C2	C	0.12112	-0.49188	-0.48171
C3	C	0.09173	-0.60254	-0.48115
N4	N	0.05219	-0.46519	-0.48208
C5	C	0.25693	-0.30807	-0.48154
H6	H	0.12317	-0.61938	-0.25838
H7	H	0.12353	-0.61947	-0.70287
H8	H	0.00862	-0.65488	-0.48179
H9	H	0.19749	-0.28902	-0.4815

Table S14. Atomic coordinates. *P6* space group coordinates of TKBPB-Ketazine-COF. $a = b = 44.7424 \text{ \AA}$, $c = 3.7100 \text{ \AA}$, $\alpha = \beta = 90^\circ$, and $\gamma = 120^\circ$.

Atom Name	Atom	x	y	z
C1	C	0.23711	-0.36547	-0.68327
C2	C	0.20121	-0.38613	-0.67405
C3	C	0.18564	-0.41921	-0.52116
C4	C	0.20666	-0.43109	-0.37352
C5	C	0.24253	-0.41011	-0.3734
C6	C	0.25845	-0.37696	-0.53024
C7	C	0.1268	-0.42719	-0.41595
C8	C	0.09093	-0.44802	-0.40366
C9	C	0.07501	-0.48334	-0.48895
C10	C	0.0959	-0.49717	-0.59132
C11	C	0.13178	-0.47636	-0.60152
C12	C	0.14758	-0.4412	-0.51322
C13	C	0.29693	-0.35454	-0.53286
C14	C	0.03682	-0.50529	-0.47505
C15	C	0.02013	-0.54378	-0.41996
N16	N	0.0181	-0.49063	-0.50141
C17	C	0.31244	-0.31837	-0.53314
H18	H	0.24791	-0.34087	-0.81957
H19	H	0.1856	-0.37661	-0.79641
H20	H	0.19527	-0.45623	-0.24677
H21	H	0.25751	-0.41972	-0.23976
H22	H	0.13835	-0.40023	-0.34288
H23	H	0.07567	-0.43653	-0.32241
H24	H	0.08459	-0.52397	-0.66919
H25	H	0.1472	-0.48763	-0.68429
H26	H	0.0375	-0.55017	-0.26668
H27	H	0.0142	-0.55722	-0.68201
H28	H	-0.00422	-0.55389	-0.26458
H29	H	0.2963	-0.3068	-0.53281

Table S15. Porosity. Porosity of COFs.

COF	BET surface area (m ² g ⁻¹)	Pore size (Å)	Pore volume (cm ³ g ⁻¹)
TKB-Ketazine- COF	340	11.3	0.1220
TKBPB-Ketazine- COF	703	28.7	0.3497

Table S16. Luminescence. Light-emitting properties of COFs.

	Solid state			
	Abs (nm)	Emission (nm)	QY (%)	Lifetime (ns)
TKB-Ketazine-COF	364	481	0.3±0.1	1.28
TKB monomer	255,300,329	452		
Wavelength Difference	35	29		
TKBPB-Ketazine-COF	394	483	0.5±0.1	1.69
TKBPB monomer	349	445		
Wavelength Difference	45	38		
	THF			
	Abs (nm)	Emission (nm)	QY (%)	Lifetime (ns)
TKB-Ketazine-COF	324	450	15.5±1.5	1.74
TKB monomer	292	437		
Wavelength Difference	32	13		
TKBPB-Ketazine-COF	415	457	14.4±0.8	1.67
TKBPB monomer	308	444		
Wavelength Difference	107	13		
	DMF			
	Abs (nm)	Emission (nm)	QY (%)	Lifetime (ns)
TKB-Ketazine-COF	326	464	8.3±0.9	5.30
TKB monomer	294	445		
Wavelength Difference	32	19		
TKBPB-Ketazine-COF	408	452	11.3±1.2	9.24
TKBPB monomer	313	443		
Wavelength Difference	95	9		
	NMP			
	Abs (nm)	Emission (nm)	QY (%)	Lifetime (ns)
TKB-Ketazine-COF	328	465	17.5±0.4	7.75
TKB monomer	295	447		
Wavelength Difference	33	18		
TKBPB-Ketazine-COF	420	448	11.7±1.6	9.14
TKBPB monomer	316	456		
Wavelength Difference	104	-8		
	DCM			
	Abs (nm)	Emission (nm)	QY (%)	Lifetime (ns)
TKB-Ketazine-COF	319	458	4.8±1.3	2.04
TKB monomer	293	440		
Wavelength Difference	26	18		
TKBPB-Ketazine-COF	417	447	5.8±1.1	1.82
TKBPB monomer	309	452		
Wavelength Difference	108	-5		

Table S17. Blue light-emitting COFs. Comparison with blue light-emitting COFs.

COFs	State	λ_{ex} (nm)	λ_{em} (nm)	ϕ (%)	τ (ns)	Reference
TKB-Ketazine-COF	solid	364	410	0.3($\pm$ 0.1)	1.28	This work
	THF	324	450	15.5($\pm$ 1.5)	1.74	
	DMF	326	464	8.3($\pm$ 0.9)	5.30	
	NMP	328	465	17.5($\pm$ 0.4)	7.75	
	DCM	319	458	4.8($\pm$ 1.3)	2.04	
TKBPB-Ketazine-COF	solid	394	483	0.5($\pm$ 0.1)	1.69	
	THF	415	457	14.4($\pm$ 0.8)	1.67	
	DMF	408	452	11.3($\pm$ 1.2)	9.24	
	NMP	420	448	11.7($\pm$ 1.6)	9.14	
	DCM	417	447	5.8($\pm$ 1.1)	1.82	
TP-COF	solid	376, 417	474	-	-	S2
PPy-COF	solid	414	484	-	-	S3
TfpBDH-CONs	IPA	365	447	-	0.75 ns	S4
COF-LZU8	solid	390	460	3.5	-	S5
	acetonitrile	390	~ 450		-	
	THF	390	~ 450			
	DMF	390	~ 450			

	ethanol	390	~ 450			
TPE-Ph COF	solid	390	500	32	0.7 ns	S6
Py-DBA-COF 2	solid	365	483	-	-	S7
PI-COF 202	acetonitrile	370	462.1	38	-	S8
Py-TPE-COF	THF	365	462	21.1	-	S9
	acetone	365	469	18.0	-	
	ACN	365	474	13.7	-	
	DMF	365	493	15.7	-	
NUS-31	acetonitrile	365	464	3.36	-	S10
NUS-32	acetonitrile	365	465	2.41	-	
NUS-33	acetonitrile	365	470	0.85	-	
TFPPy-CHYD COF	DMF	325	463	13.6	2.5	S11
BCTB-PD COF	pyridine	365	462	12.8	-	S12
	acetone	365	469	13.5	-	

Table S18 Energy levels. Redox potentials and energy levels of COFs.

	Reduction Potential (V)	Oxidation Potential (V)	HOMO (eV)	LUMO (eV)	$E_{g, \text{opt}}$ (eV)	$E_{g, \text{electrochem}}$ (eV)
TKB-Ketazine- COF	−1.20	0.43	−5.02	−3.39	2.74	1.63
TKBPB-Ketazine- COF	−1.10	0.44	−5.03	−3.49	2.64	1.54

Section 4. Supporting Figures

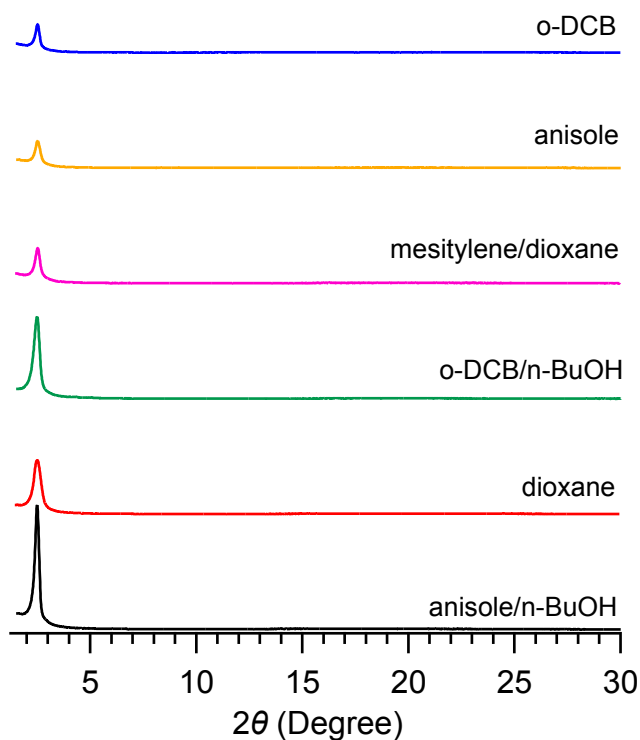


Figure S1. Effect of Solvent. PXRD patterns of TKBPB-Ketazine-COF synthesized in different solvent conditions (Table S2).

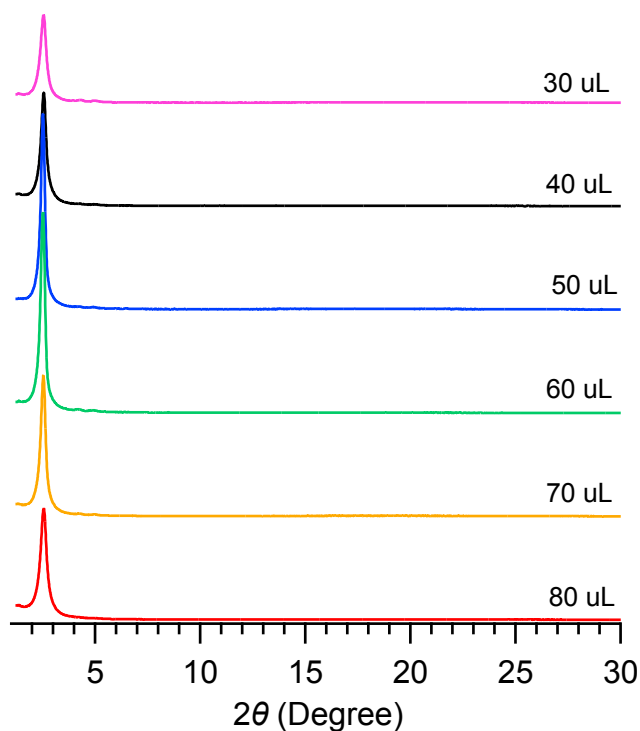


Figure S2. Effect of Amount of Acid. PXRD patterns of TKBPB-Ketazine-COF catalyzed by different amounts of pure acetic acid (Table S3).

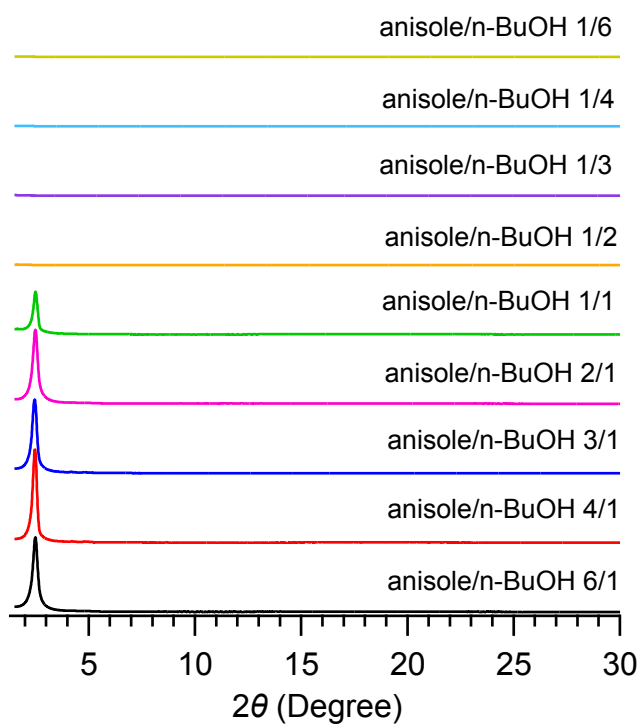


Figure S3. Effect of Ratio of Solvent Combinations. PXRD patterns of TKBPB-Ketazine-COF synthesized in different ratio of solvents (Table S4).

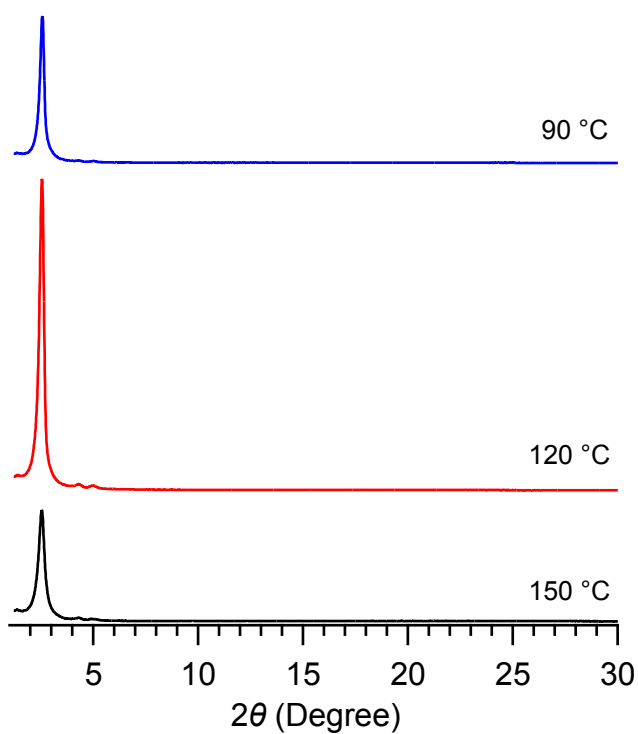


Figure S4. Effect of Temperatures. PXRD patterns of TKBPB-Ketazine-COF synthesized under different temperatures (Table S5).

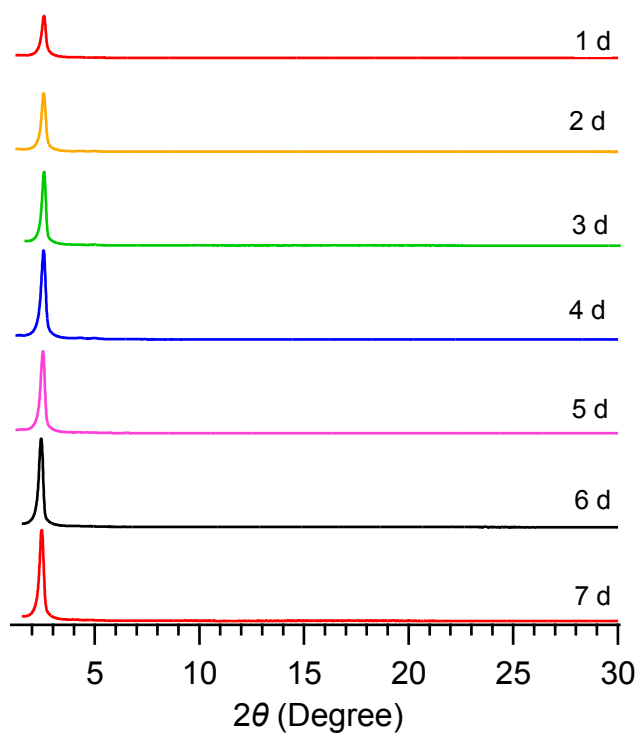


Figure S5. Effect of Reaction Time. PXRD patterns of TKBPB-Ketazine-COF with different reaction times (Table S6).

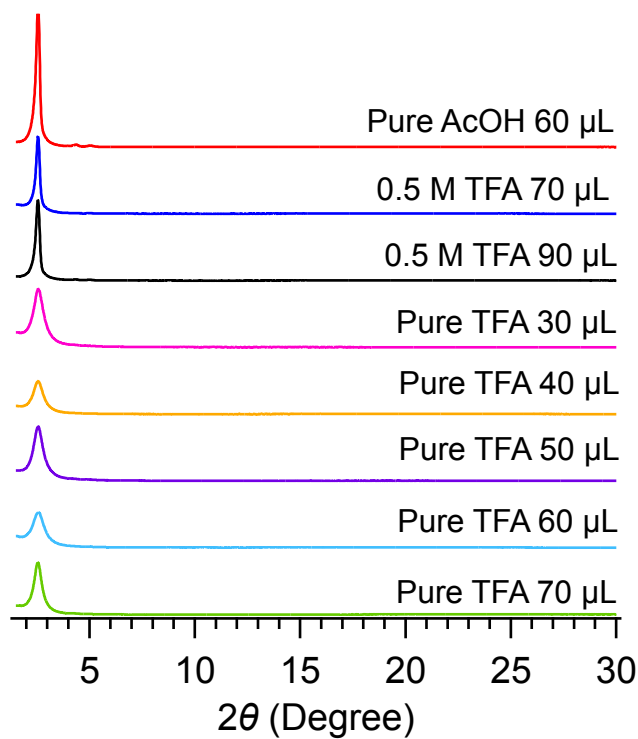


Figure S6. Effect of trifluoroacetic acid catalyst. PXRD patterns of TKBPB-Ketazine-COF with trifluoroacetic acid catalyst.

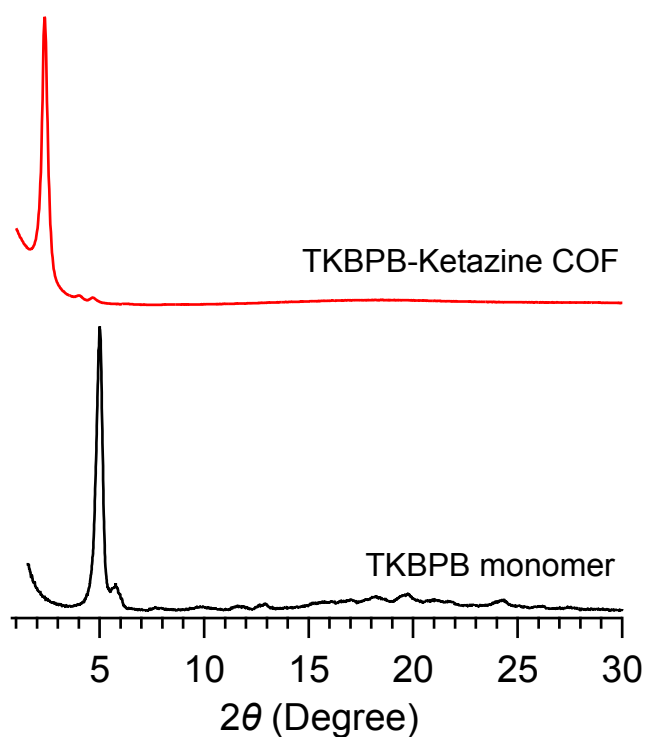


Figure S7. Comparison with monomer. PXRD patterns of TKB-Ketazine-COF (red) and TKBPB monomer (black).

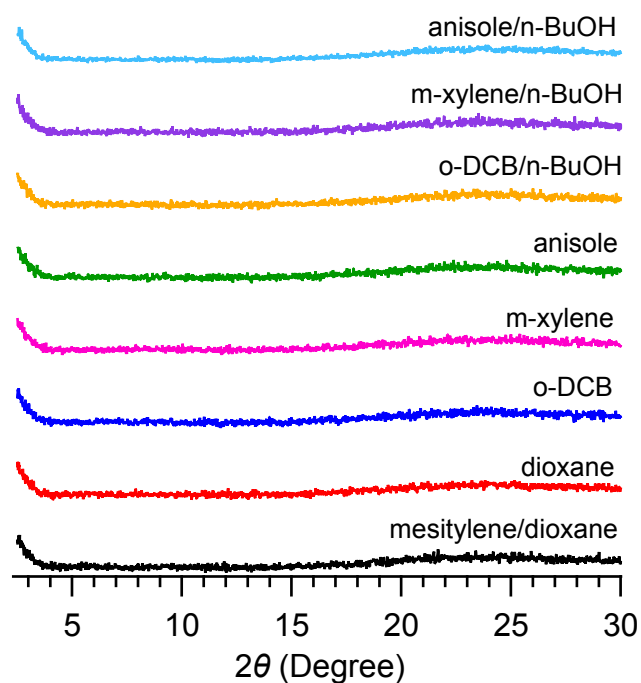


Figure S8. Effect of Solvent. PXRD patterns of TKB-Ketazine-COF (gel) synthesized in different solvent conditions (Table S8).

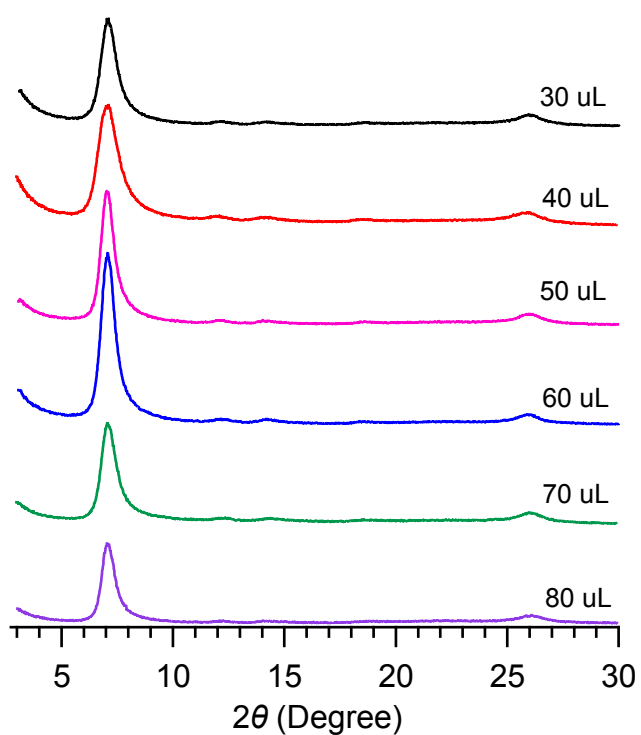


Figure S9. Effect of Amount of Acid. PXRD patterns of TKB-Ketazine-COF catalyzed by different amounts of TFA (Table S9).

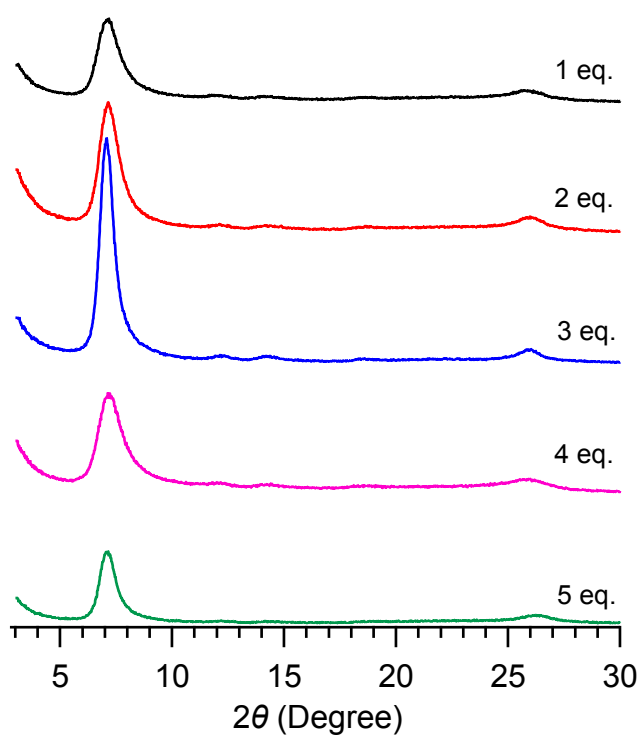


Figure S10. Effect of Amount of Modulator. PXRD patterns of TKB-Ketazine-COF modulated by different amounts of acetophenone (Table S10).

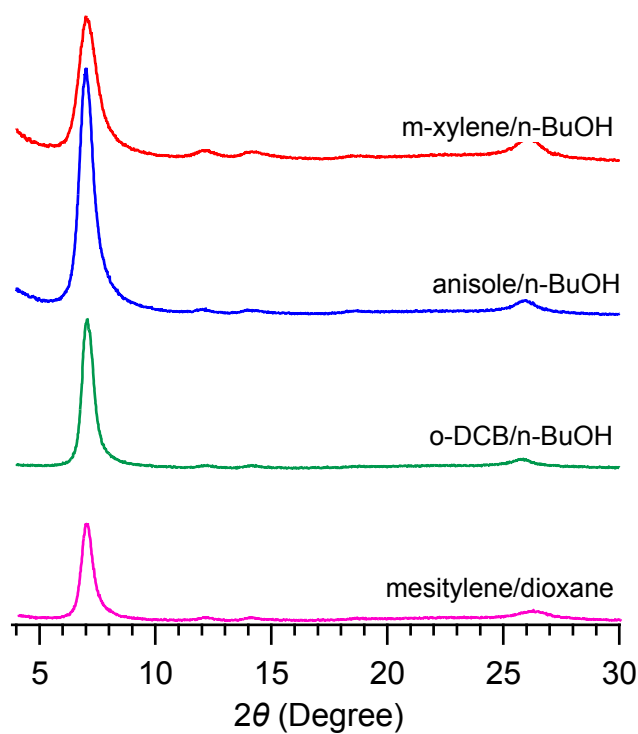


Figure S11. Effect of Solvents. PXRD patterns of TKB-Ketazine-COF synthesized in different solvent conditions (Table S11).

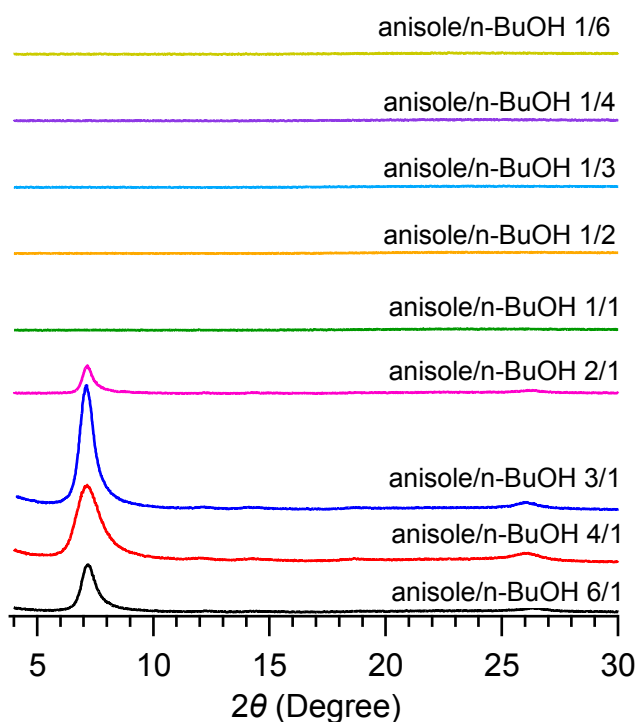


Figure S12. Effect of Ratio of Solvent Combinations. PXRD patterns of TKB-Ketazine-COF synthesized in different ratio of solvents (Table S11).

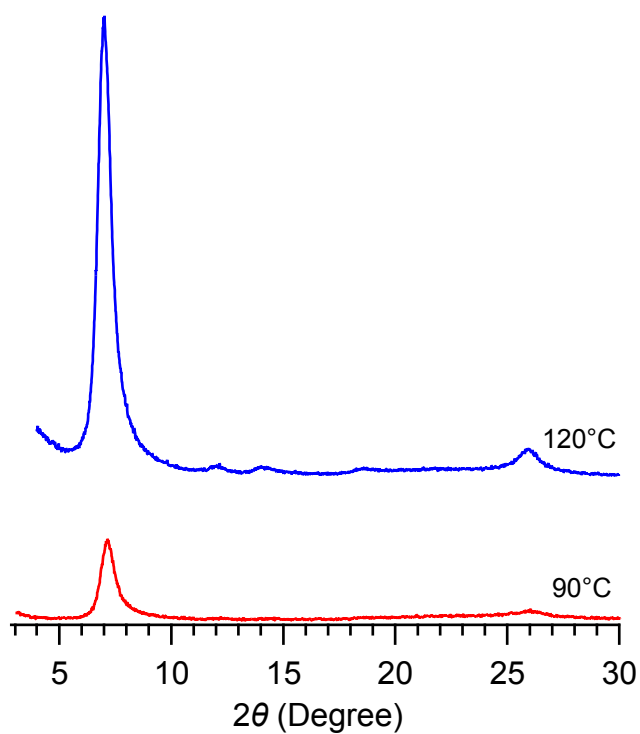


Figure S13. Effect of Temperatures. PXRD patterns of TKB-Ketazine-COF synthesized under different temperatures (Table S11).

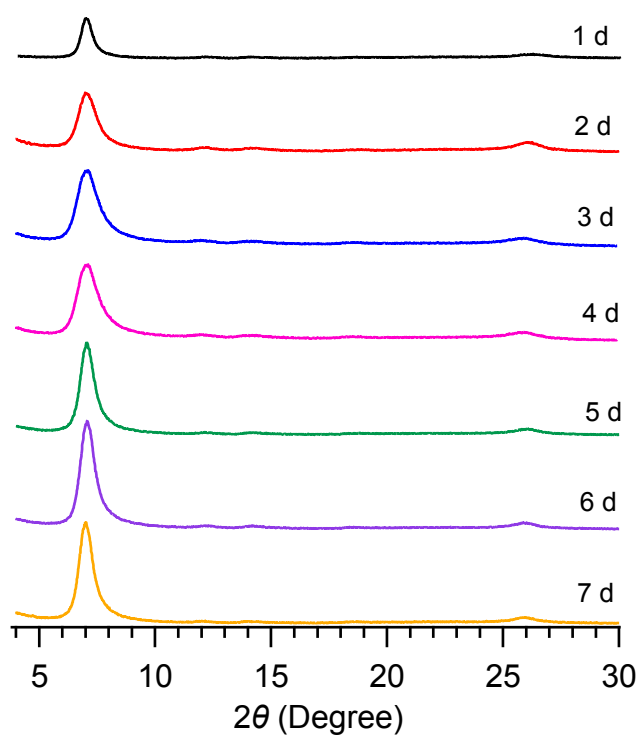


Figure 14. Effect of Reaction Time. PXRD patterns of TKB-Ketazine-COF with different reaction times (Table S11).

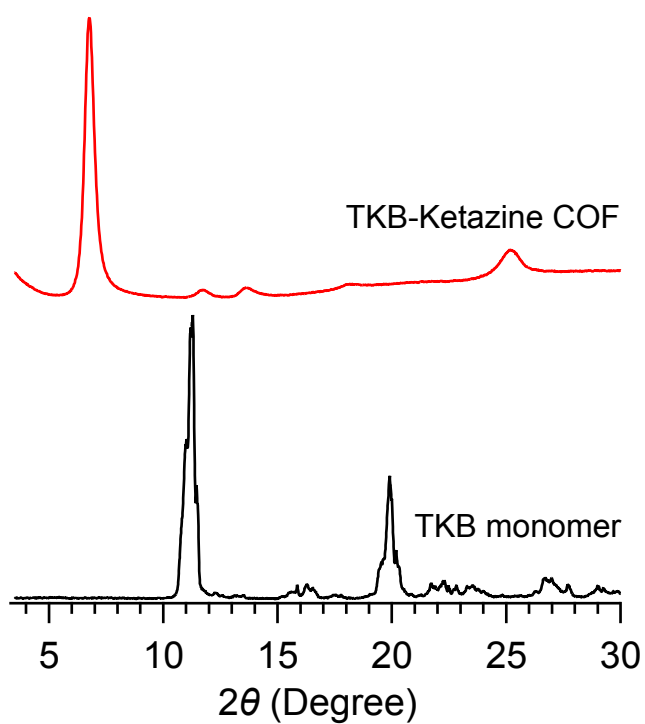


Figure S15. Comparison with monomer. PXRD patterns of TKB-Ketazine-COF (red) and TKB monomer (black).

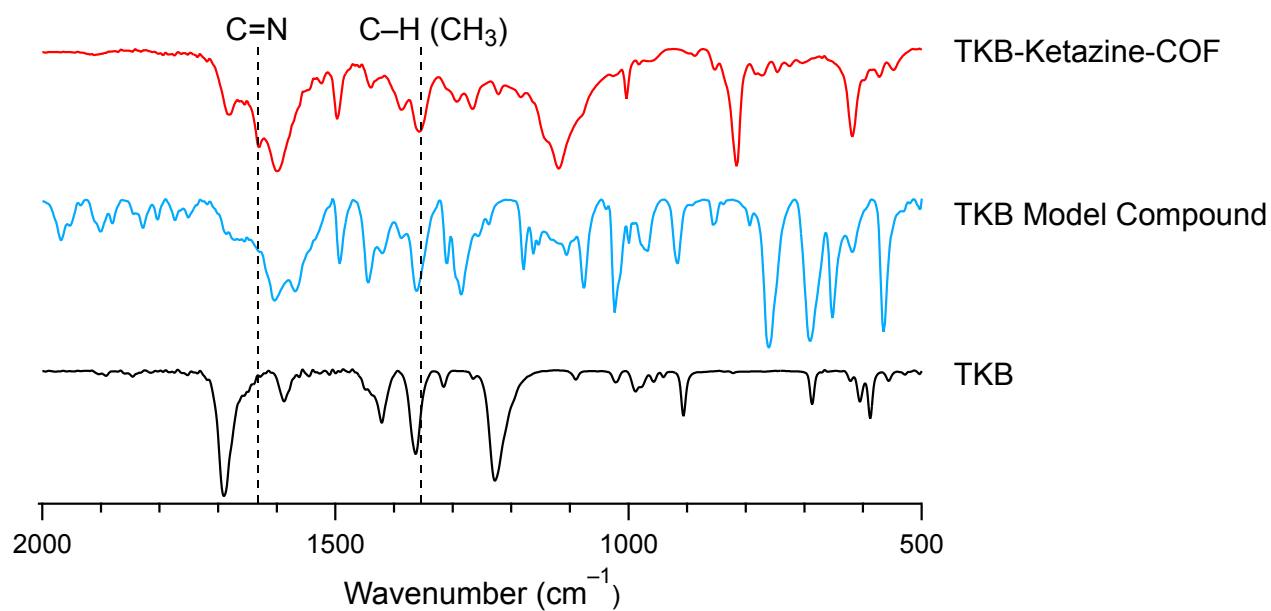


Figure S16. FT IR spectra. FT IR spectra of TKB-Ketazine-COF (red line) and TKB monomer (black line).

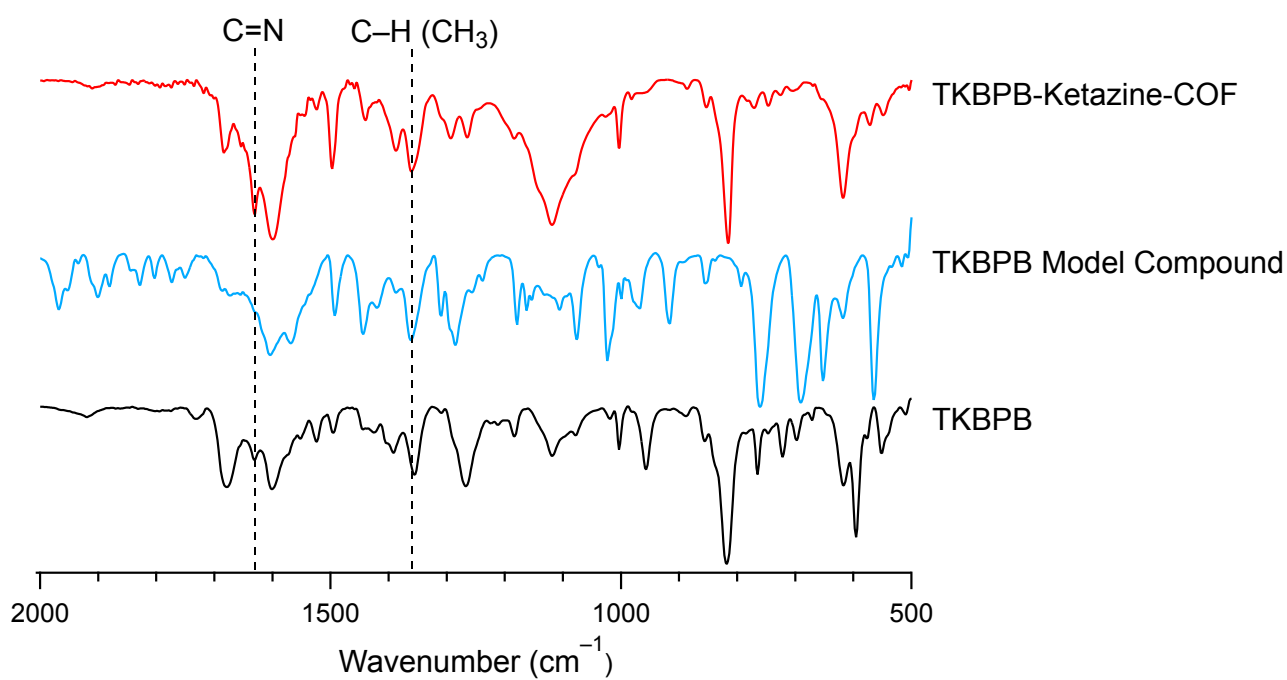


Figure S17. FT IR spectra. FT IR spectra of TKBPB-Ketazine-COF (red line) and TKBPB monomer (black line).

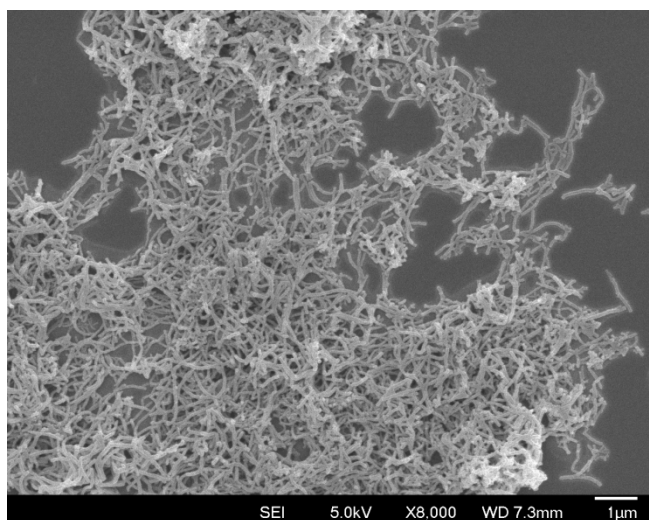


Figure S18. SEM image. SEM images of TKB-Ketazine-COF.

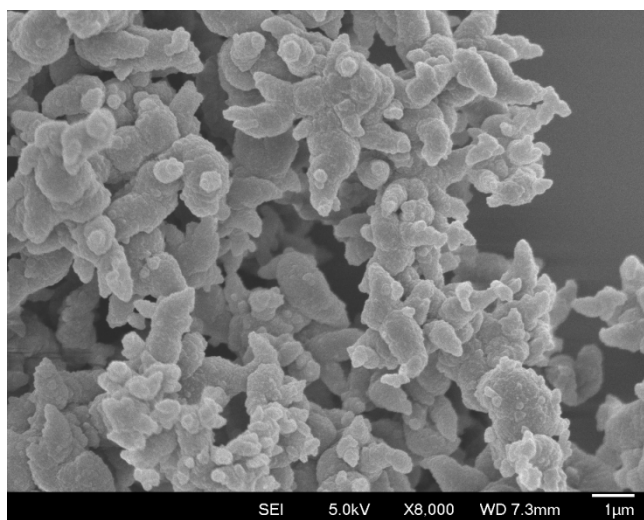


Figure S19. SEM image. SEM images of TKBPB-Ketazine-COF.

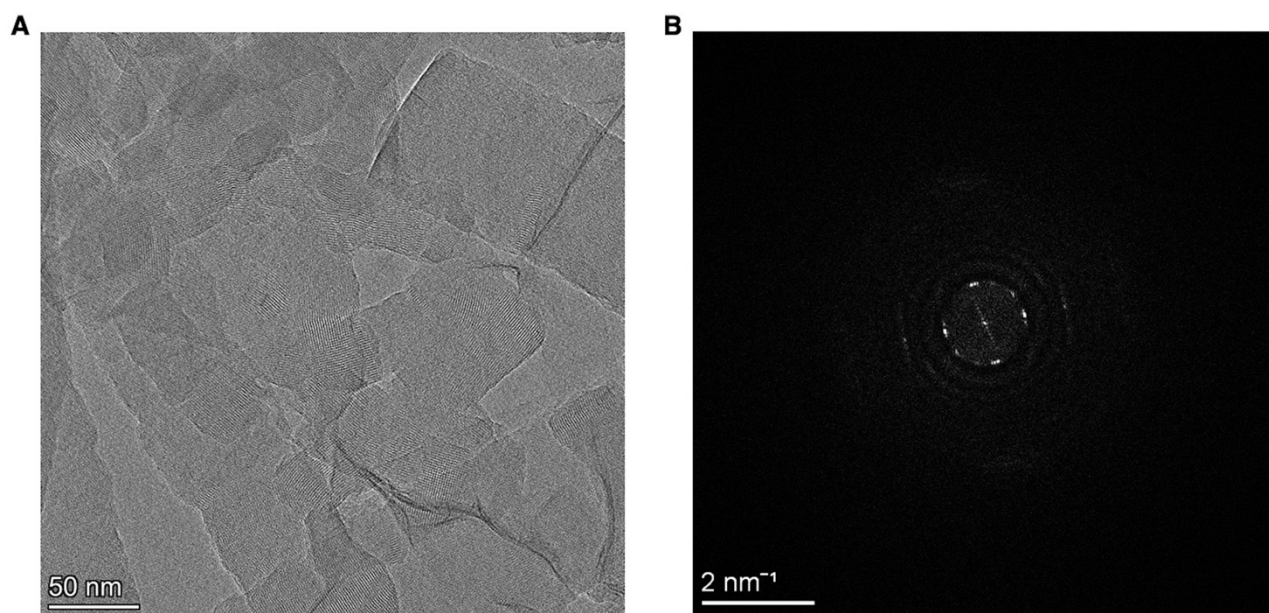


Figure S20. HR TEM image. (A) HR TEM image of TKB-Ketazine-COF. (B) FFT file.

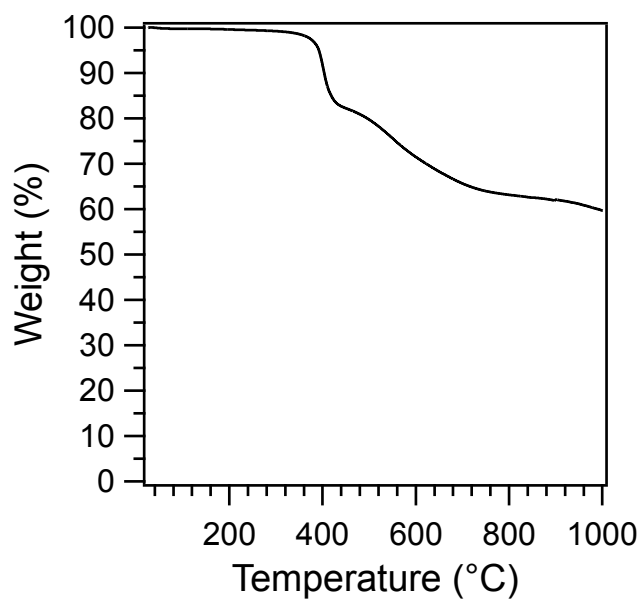


Figure S21. TGA curves. TGA curves of TKB-Ketazine-COF.

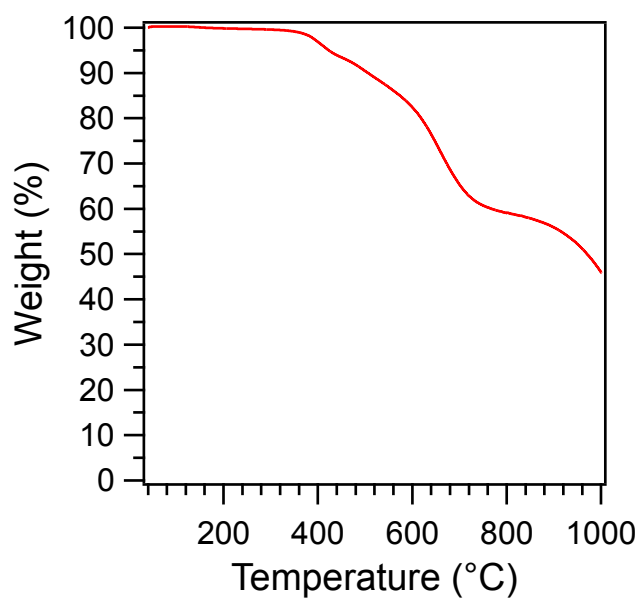


Figure S22. TGA curves. TGA curves of TKBPB-Ketazine-COF.

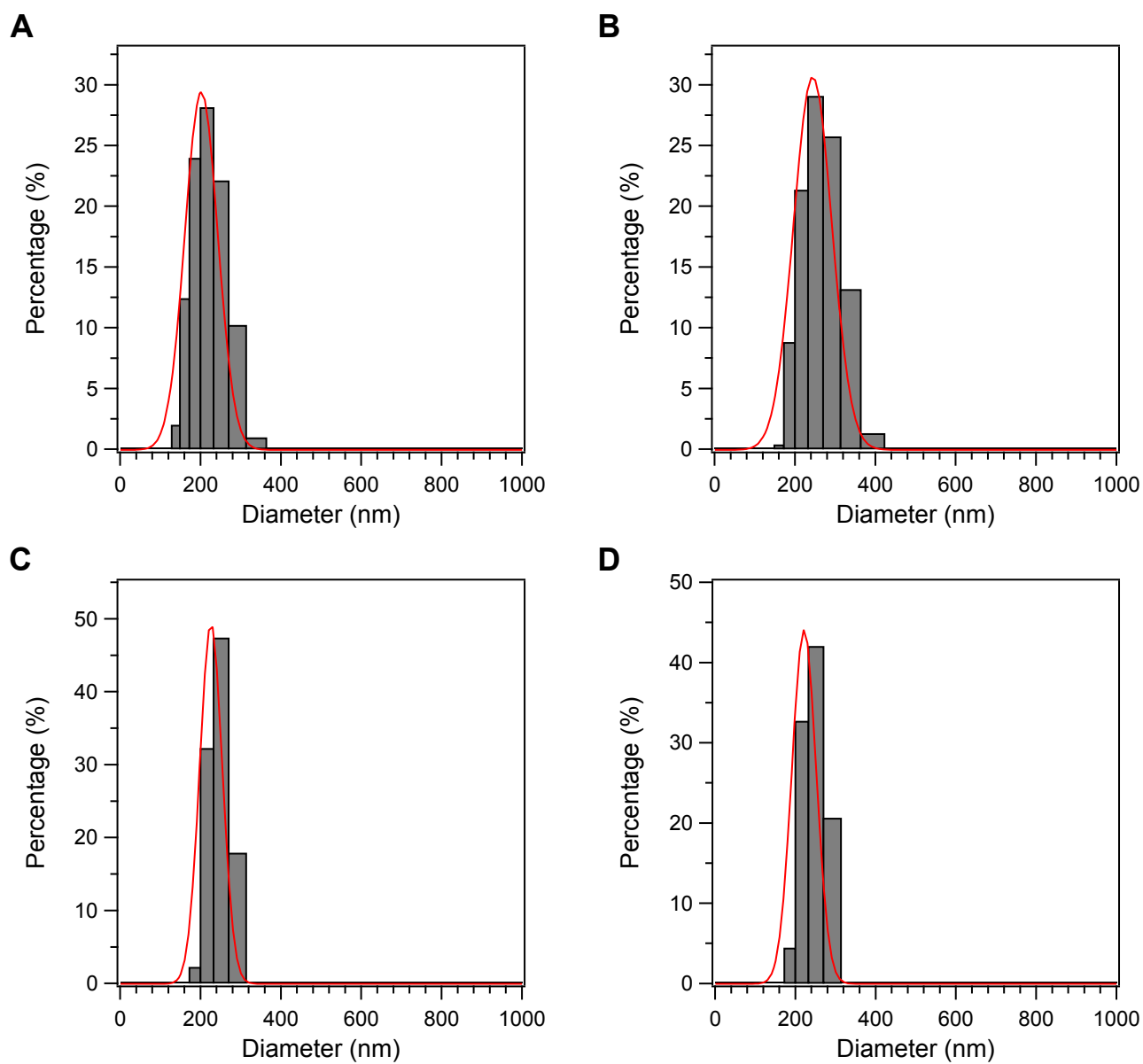


Figure S23. DLS results. DLS results of TKB-Ketazine-COF dispersed in (A) DMF, (B) THF, (C) DCM and (D) NMP.

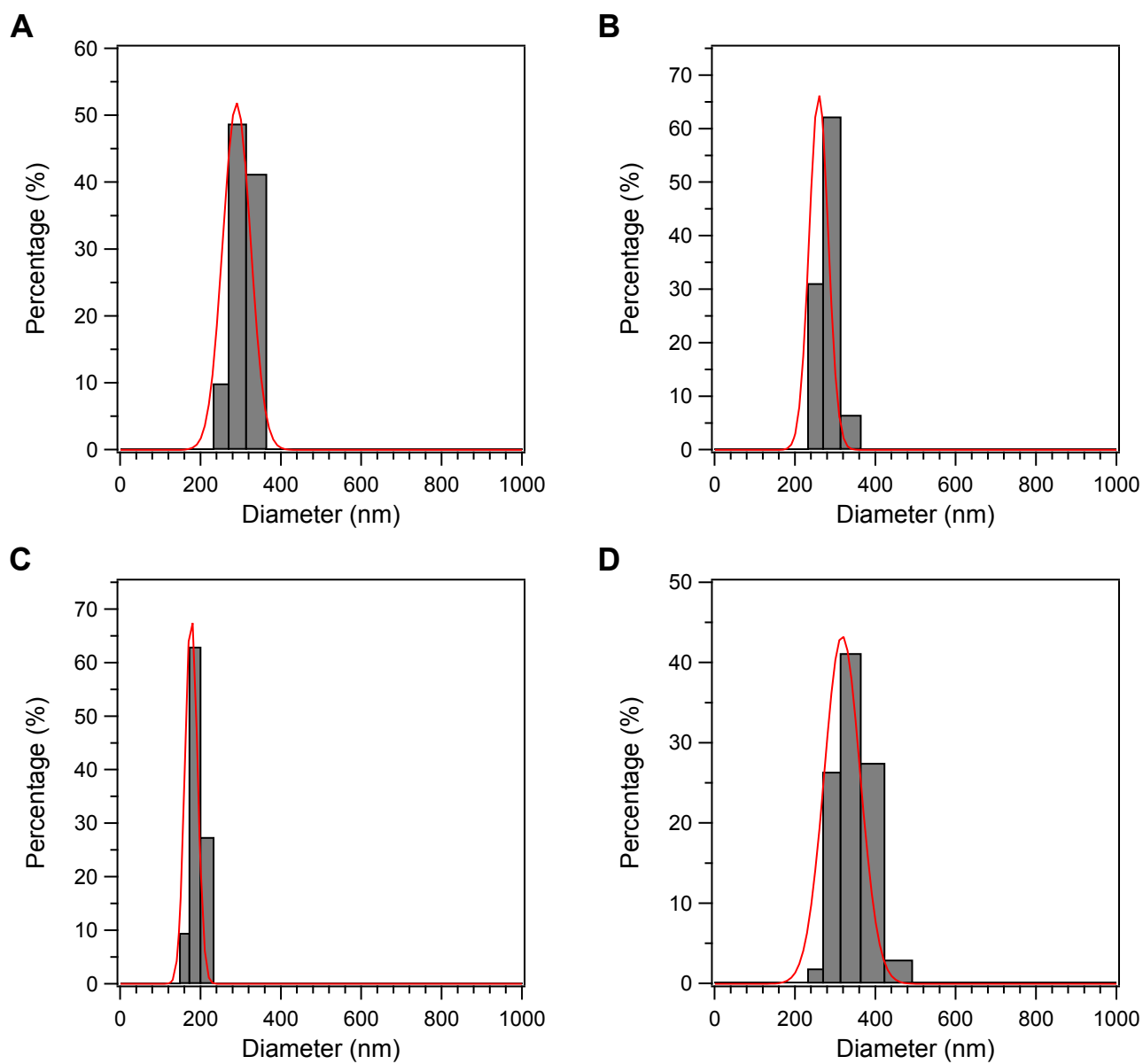


Figure S24. DLS results. DLS results of TKBPB-Ketazine-COF dispersed in (A) DMF, (B) THF, (C) DCM and (D) NMP.

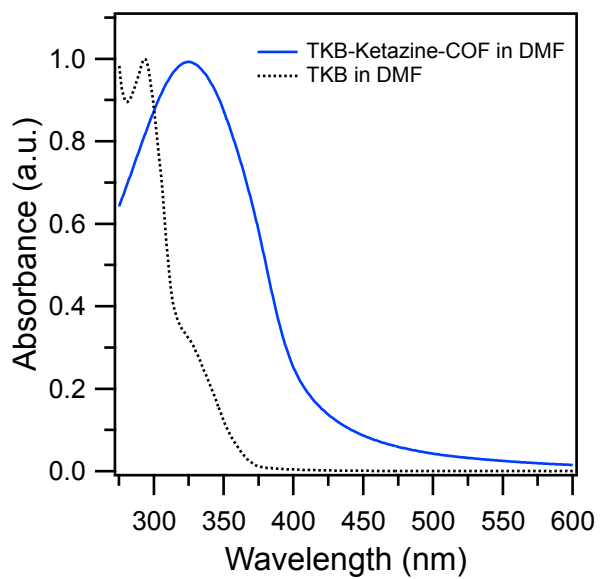


Figure S25. Absorption spectra. Absorption spectra of TKB-Ketazine-COF (blue solid line) and TKB monomer (blue dashed line) in DMF.

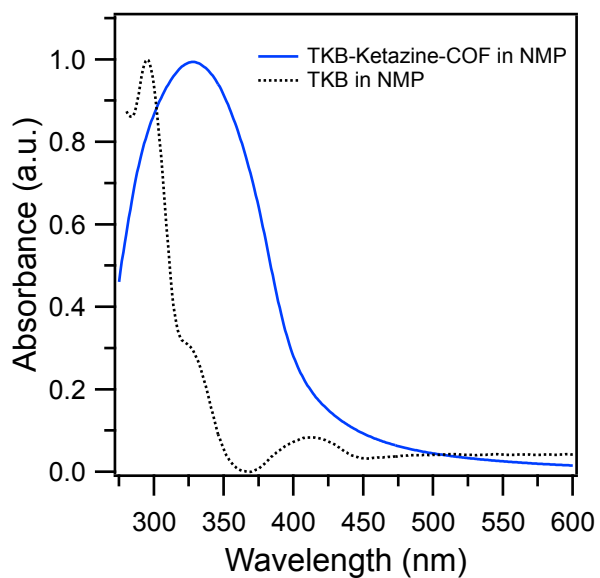


Figure S26. Absorption spectra. Absorption spectra of TKB-Ketazine-COF (blue solid line) and TKB monomer (blue dashed line) in NMP.

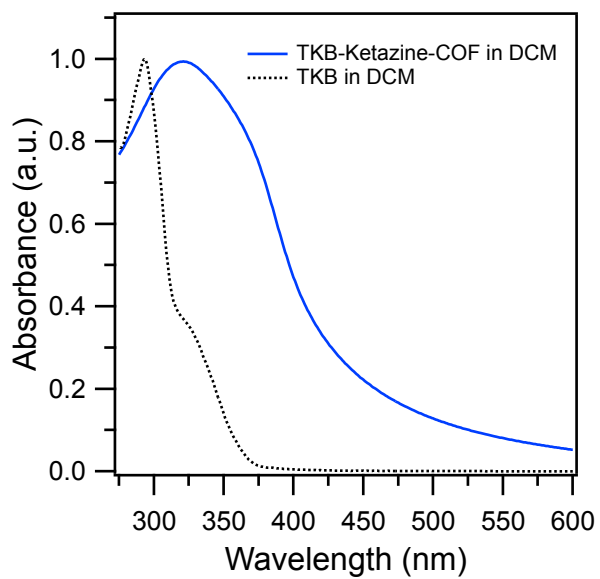


Figure S27. Absorption spectra. Absorption spectra of TKB-Ketazine-COF (blue solid line) and TKB monomer (blue dashed line) in DCM.

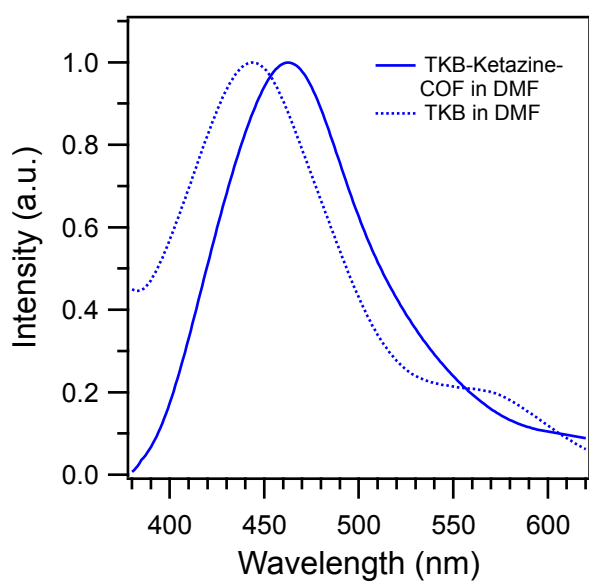


Figure S28. Fluorescence spectra. Fluorescence spectra of TKB-Ketazine-COF (blue solid line) and TKB monomer (blue dashed line) in DMF.

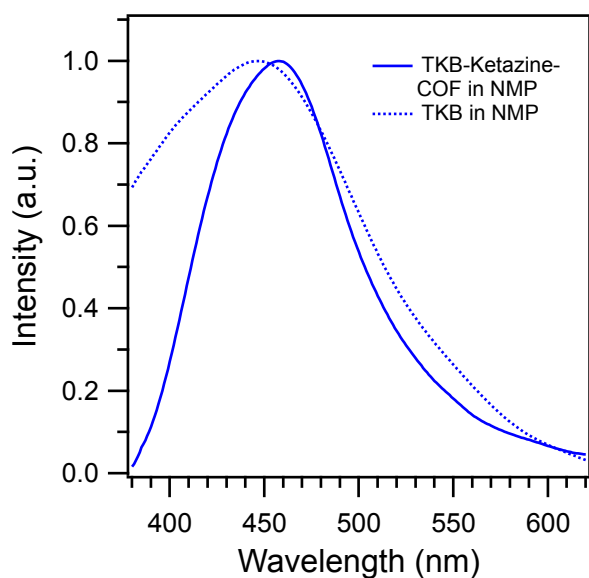


Figure S29. Fluorescence spectra. Fluorescence spectra of TKB-Ketazine-COF (blue solid line) and TKB monomer (blue dashed line) in NMP.

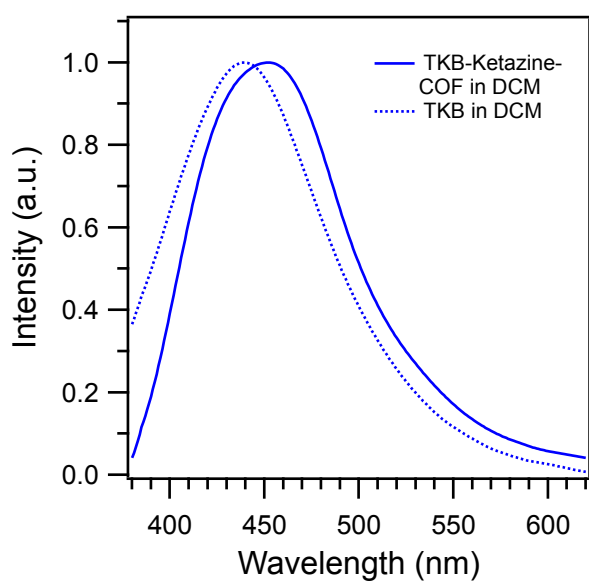


Figure S30. Fluorescence spectra. Fluorescence spectra of TKB-Ketazine-COF (blue solid line) and TKB monomer (blue dashed line) in DCM.

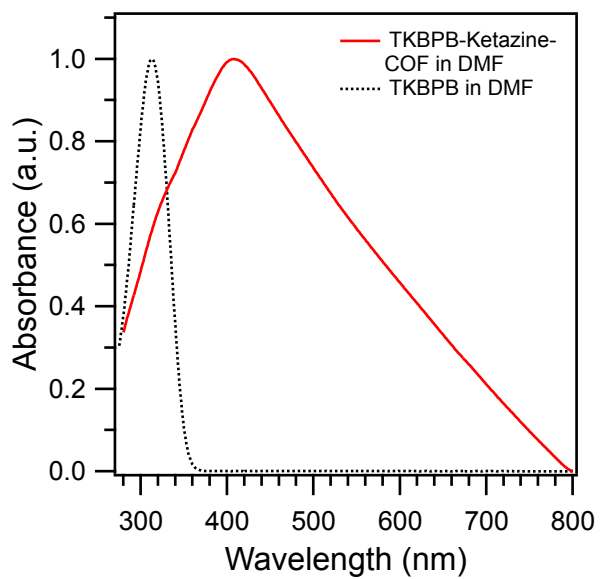


Figure S31. Absorption spectra. Absorption spectra of TKBPB-Ketazine-COF (red solid line) and TKBPB monomer (red dashed line) in DMF.

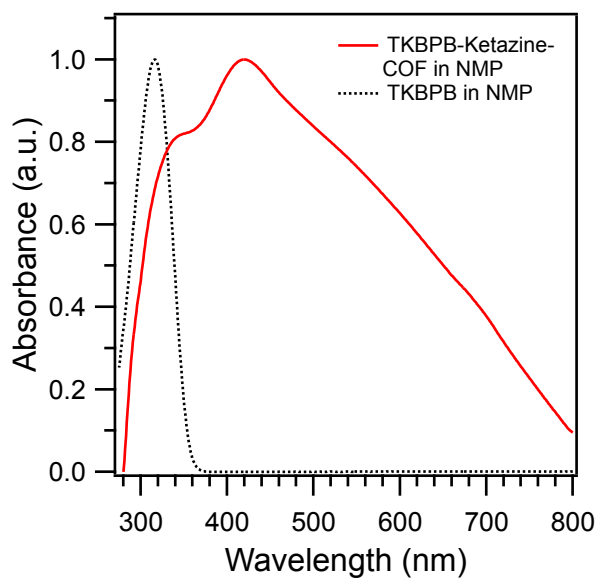


Figure S32. Absorption spectra. Absorption spectra of TKBPB-Ketazine-COF (red solid line) and TKBPB monomer (red dashed line) in NMP.

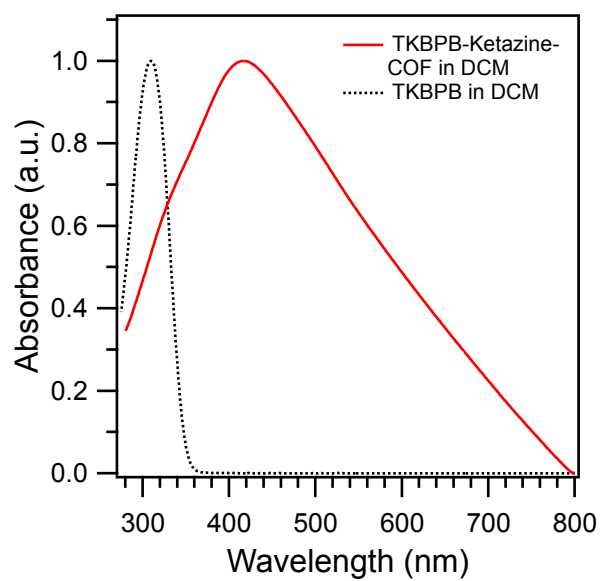


Figure S33. Absorption spectra. Absorption spectra of TKBPB-Ketazine-COF (red solid line) and TKBPB monomer (red dashed line) in DCM.

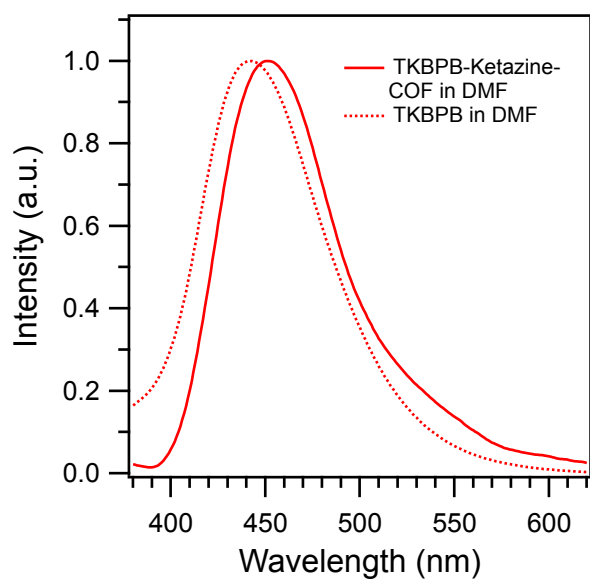


Figure S34. Fluorescence spectra. Fluorescence spectra of TKBPB-Ketazine-COF (red solid line) and TKBPB monomer (red dashed line) in DMF.

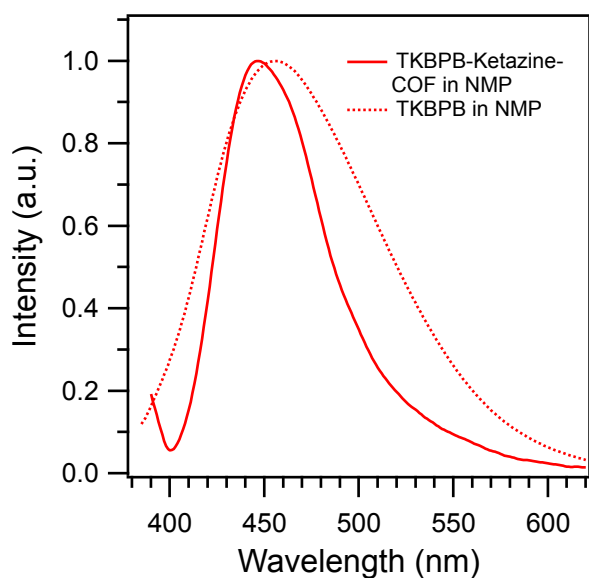


Figure S35. Fluorescence spectra. Fluorescence spectra of TKBPB-Ketazine-COF (red solid line) and TKBPB monomer (red dashed line) in NMP.

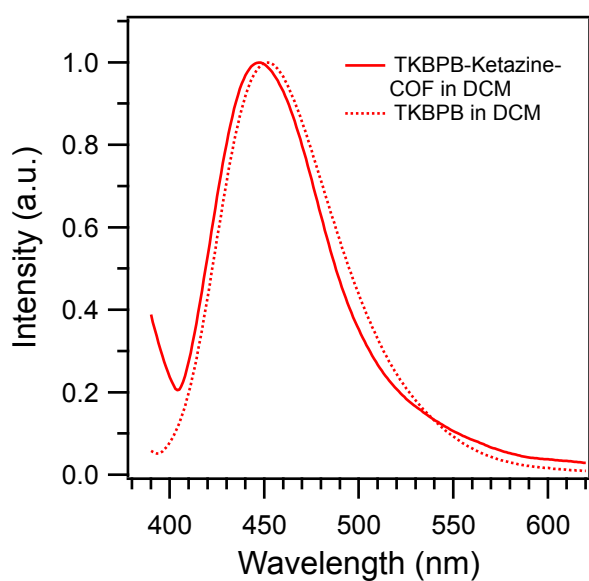


Figure S36. Fluorescence spectra. Fluorescence spectra of TKBPB-Ketazine-COF (red solid line) and TKBPB monomer (red dashed line) in DCM.

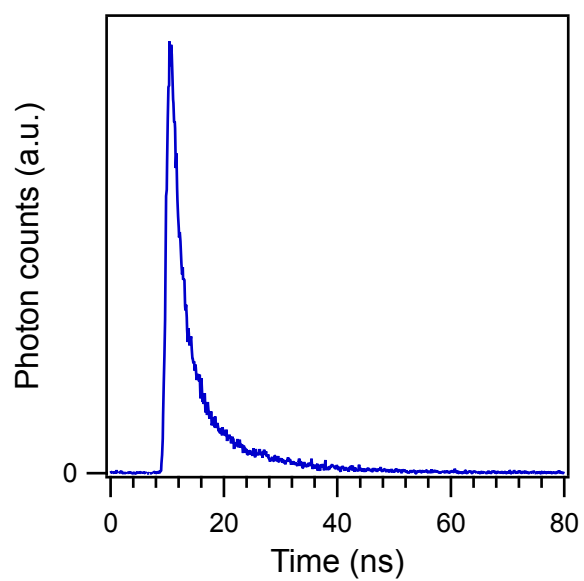


Figure S37. Transient decay profiles. Transient decay profiles of TKB-Ketazine-COF dispersed in DMF (Photon counts are 1000).

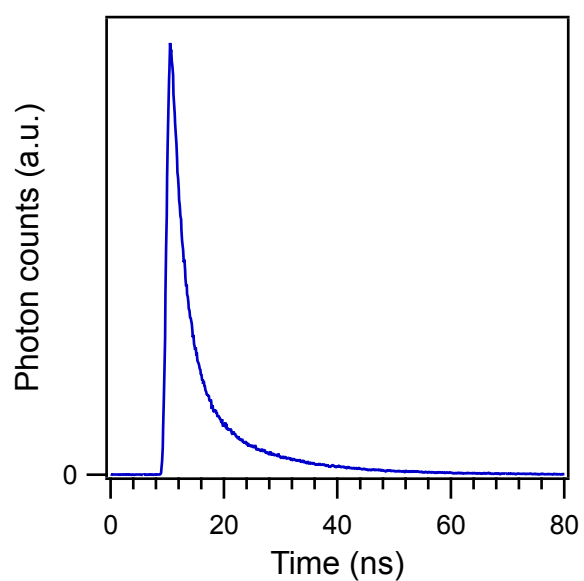


Figure S38. Transient decay profiles. Transient decay profiles of TKB-Ketazine-COF dispersed in NMP (Photon counts are 1000).

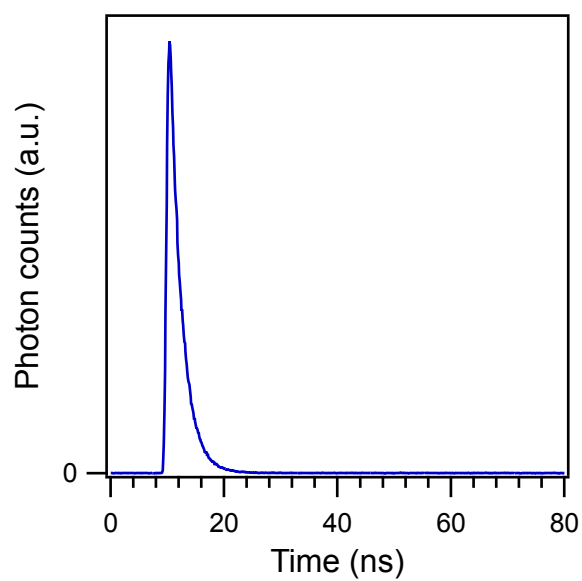


Figure S39. Transient decay profiles. Transient decay profiles of TKB-Ketazine-COF dispersed in DCM (Photon counts are 1000).

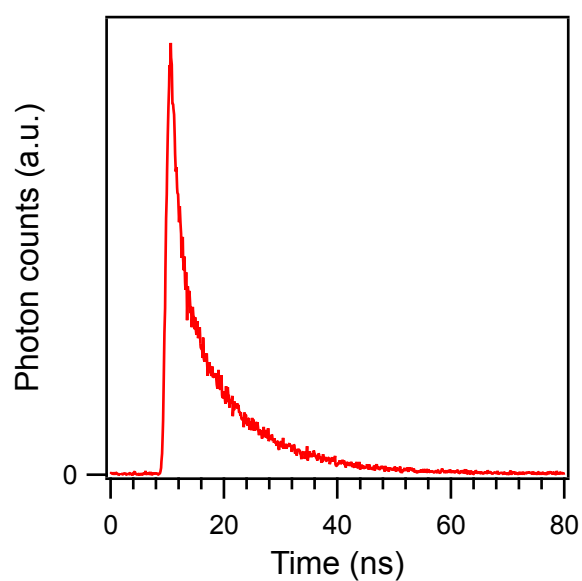


Figure S40. Transient decay profiles. Transient decay profiles of TKBPB-Ketazine-COF dispersed in DMF (Photon counts are 1000).

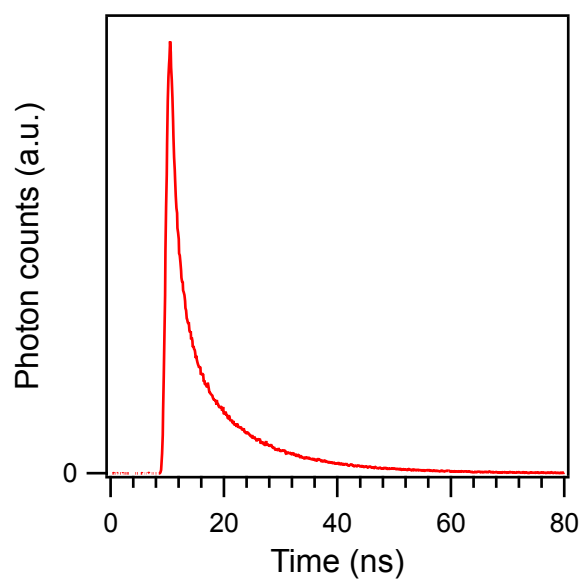


Figure S41. Transient decay profiles. Transient decay profiles of TKBPB-Ketazine-COF dispersed in NMP (Photon counts are 1000).

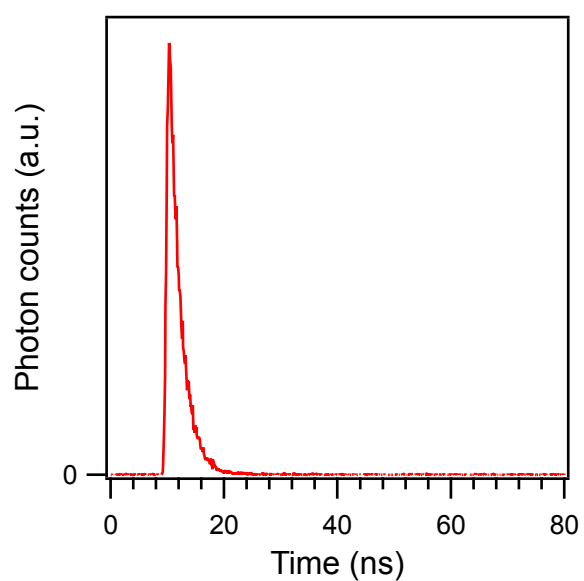


Figure S42. Transient decay profiles. Transient decay profiles of TKBPB-Ketazine-COF dispersed in DCM (Photon counts are 1000).

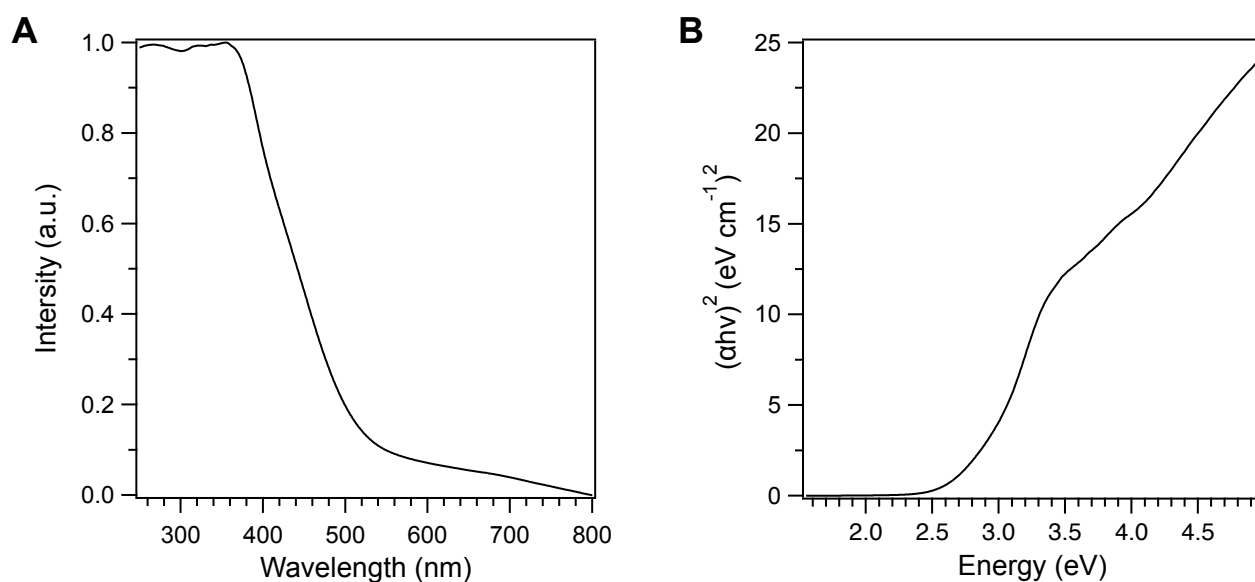


Figure S43. Optical bandgap. (A) Electronic absorption spectrum and (B) Tauc plot analysis of TKB-Ketazine-COF.

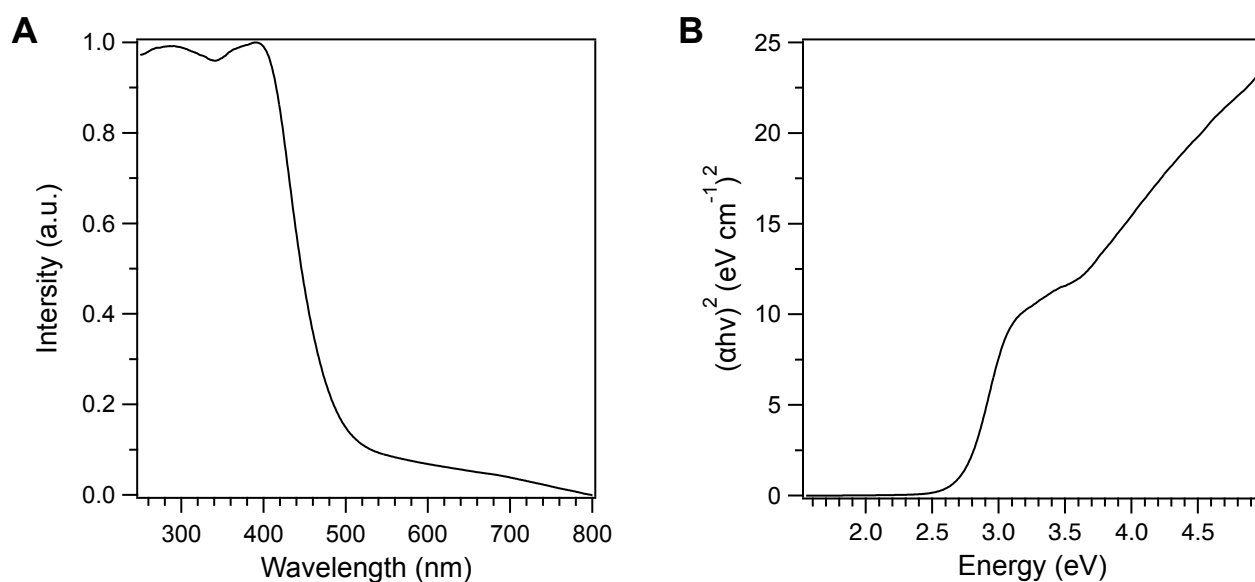


Figure S44. Optical bandgap. (A) Electronic absorption spectrum and (B) Tauc plot analysis of TKBPB-Ketazine-COF.

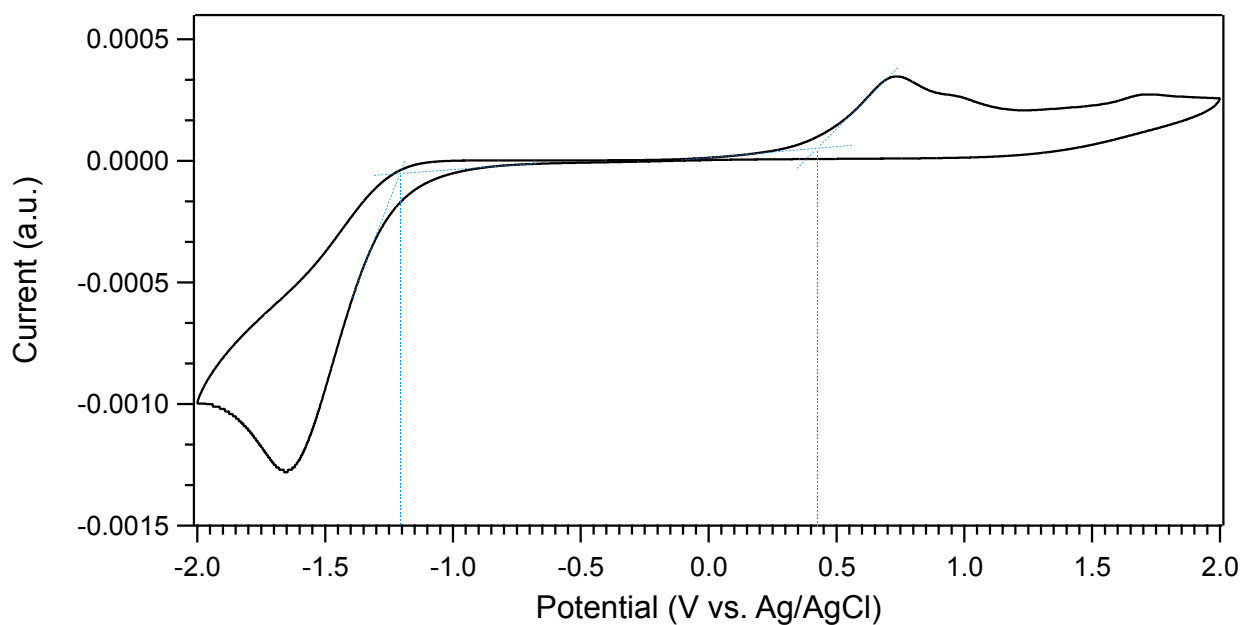


Figure S45. CV. CV curve of TKB-Ketazine-COF.

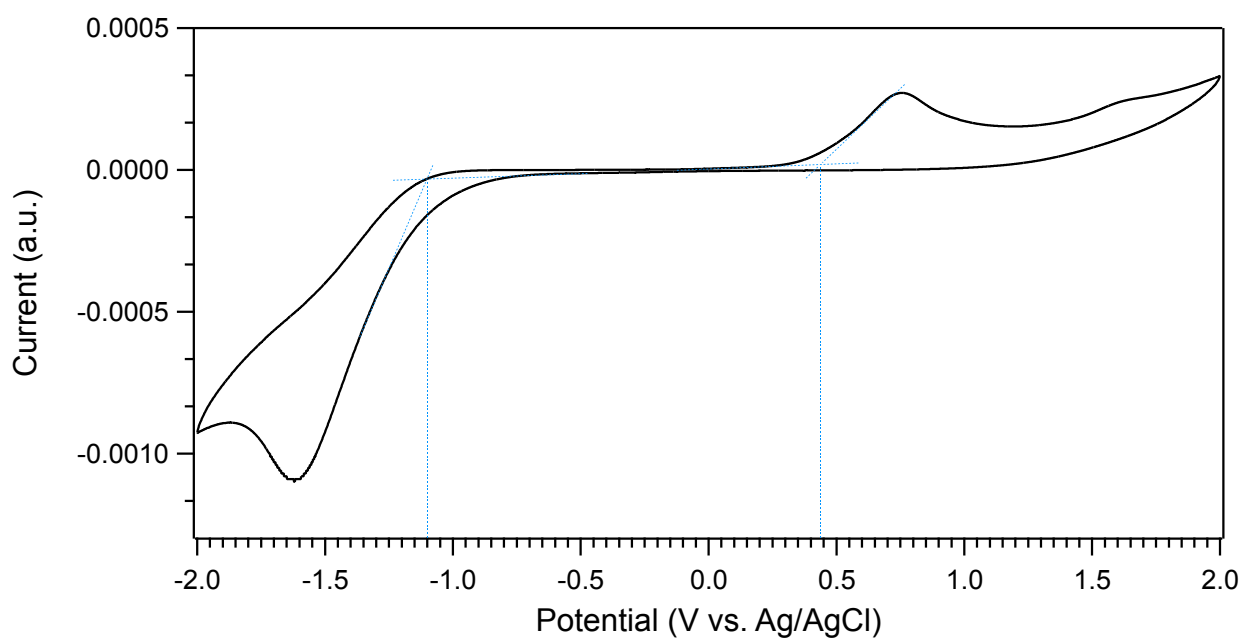


Figure S46. CV. CV curves of TKBPB-Ketazine-COF.

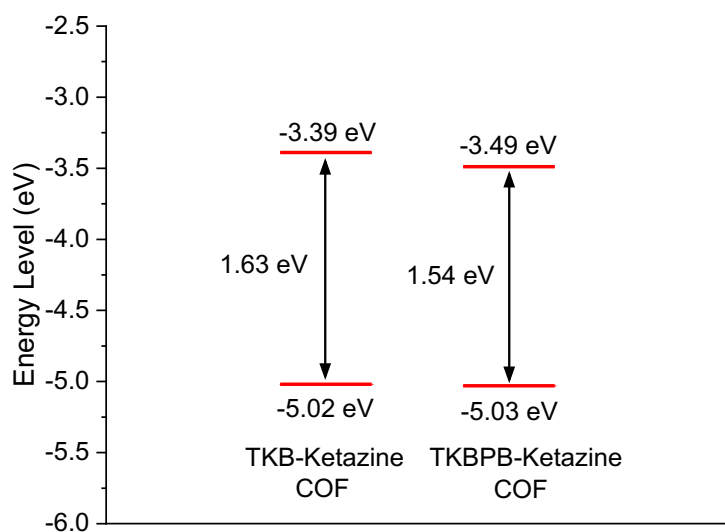


Figure S47. Bandgap structure. (A and B) Bandgap structures of (A) TKBPB-Ketazine-COF and (B) TKB-Ketazine-COF.

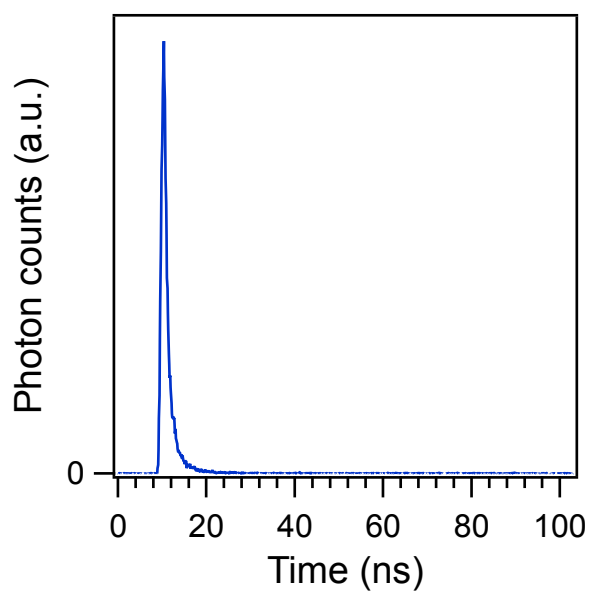


Figure S48. Transient decay profiles. Transient decay profiles of solid-state TKB-Ketazine-COF (Photon counts are 1000).

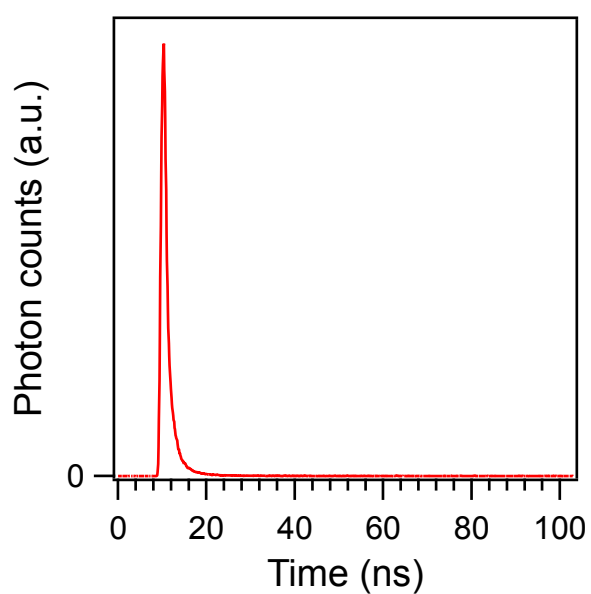


Figure S49. Transient decay profiles. Transient decay profiles of solid-state TKBPB-Ketazine-COF (Photon counts are 1000).

Section 5. Supporting References

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